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21 February 2024
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23 July 2018

DEPARTMENT OF DEFENSE

DESIGN CRITERIA STANDARD

MATERIAL AND PROCESS REQUIREMENTS
FOR AEROSPACE WEAPONS SYSTEMS

AMSC 10461

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. The purpose of this standard is to establish materials and process requirements for aerospace weapons systems.

3. Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center, Aircraft Division Lakehurst, Systems Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocComments@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

4. For guidance on the technical content of this document, contact the Commanding Officer, Naval Air Warfare Center, Aircraft Division (Code AB23000), 48066 Shaw Road, 2188 Patuxent River, MD 20670.

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1. SCOPE

1.1 Scope. This standard defines the engineering requirements, methods and procedures for the selection and use of materials and processes in the design and production of military aerospace weapons systems and associated support equipment. This document in conjunction with MIL-STD-1530, MIL-STD-1798, MIL-STD-3024, MIL-M-8856, MIL-HDBK-513 and MIL-STD-1568 provides the engineering basis for proper selection of materials and processes for reliable aerospace systems with consideration of acquisition and life cycle costs. Authority to deviate from the requirements in this standard requires approval from the cognizant engineering authority (CEA) of the procuring activity.

1.2 Applicability. This standard is applicable for use by all Department of Defense procuring activities and their respective contractors involved in the design and procurement of aerospace weapons systems and support equipment.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 4 or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59588	Rubber, Silicone
A-A-59877	Insulating Compound, Electrical, Embedding

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-PRF-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for
MIL-PRF-8184	Plastic Sheet, Acrylic, Modified
MIL-PRF-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Chemically Cured
MIL-M-8856	Missiles, Guided, Structural Integrity General Specification for

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MIL-S-9041	Sandwich Construction; Plastic Resin, Glass Fabric Base, Laminated Facings and Honeycomb Core for Aircraft Structural and Electronic Applications
MIL-PRF-23377	Primer Coatings: Epoxy, High-Solids
MIL-PRF-23586	Sealing Compound (With Accelerator), Silicone Rubber, Electrical
MIL-M-24041	Molding and Potting Compound, Chemically Cured, Polyurethane
MIL-S-25392	Sandwich Construction, Plastic Resin, Glass Fabric Base, Laminated Facings and Polyurethane Foamed-in-Place Core, for Aircraft Structural Applications
MIL-PRF-25690	Plastic, Sheet and Formed Parts, Modified Acrylic Base, Monolithic, Crack Propagation Resistant
MIL-PRF-25732	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Limited Service at 275 Degrees F (135 DEG. C)
MIL-DTL-25988	Rubber, Fluorosilicone Elastomer, Oil- and Fuel-Resistant, Sheets, Strips, Molded Parts and Extruded Shapes
MIL-DTL-25988/1	Rubber, Fluorosilicone Elastomer, Oil- and Fuel-Resistant, O-Rings, Class 1, Grade 70
MIL-DTL-25988/2	Rubber, Fluorosilicone Elastomer, Oil- and Fuel-Resistant, O-Rings, Class 3
MIL-DTL-25988/3	Rubber, Fluorosilicone Elastomer, Oil- and Fuel-Resistant, O-Rings, Grade 60
MIL-DTL-25988/4	Rubber, Fluorosilicone Elastomer, Oil- and Fuel-Resistant, O-Rings, Class 1, Grade 80
MIL-DTL-32495	Powders for Cold Spray Deposition
MIL-A-46106	Adhesive Sealants, Silicone RTV
MIL-A-46146	Adhesives-Sealants, Silicone, RTV, Noncorrosive (for use with Sensitive Metals and Equipment)
MIL-DTL-81381	Wire, Electric, Polyimide-Insulated, Copper or Copper Alloy
MIL-T-81556	Titanium and Titanium Alloys, Extruded Bars and Shapes, Aircraft Quality
MIL-DTL-83397	Rubber, Polyurethane, Castable, Humidity Resistant
MIL-PRF-83483	Anti-seize Thread Compound, Molybdenum Disulfide-Petrolatum
MIL-S-85334	Sealing Compound, Noncuring, Low Consistency, Silicone, Groove Injection, for Integral Fuel Tanks
MIL-PRF-85582	Primer Coatings: Epoxy, Waterborne

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-866	Grinding of Chrome Plated Steel and Steel Parts Heat Treated to 180,000 PSI or Over
MIL-STD-867	Temper Etch Inspection
MIL-STD-889	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-1530	Aircraft Structural Integrity Program (ASIP)

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MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems
MIL-STD-1798	Mechanical Equipment and Subsystems Integrity Program
MIL-STD-3021	Materials Deposition, Cold Spray
MIL-STD-3024	Propulsion System Integrity Program (PSIP)
MIL-STD-7179	Finishes, Coatings, and Sealants, for the Protection of Aerospace Weapons Systems and Support Equipment

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-275	Guide for Selection of Lubricants, Fluids, and Compounds for Use in Flight Vehicles and Components
MIL-HDBK-513	Air Vehicle Low Observable Integrity Program General Guidelines
MIL-HDBK-516	Airworthiness Certification Criteria
MIL-HDBK-838	Lubrication of Military Equipment
MIL-HDBK-1823	Nondestructive Evaluation System Reliability Assessment
MIL-HDBK-83377	Adhesive Bonding (Structural) for Aerospace and Other Systems, Requirements for

JOINT SERVICE SPECIFICATION GUIDES

JSSG-2006	Aircraft Structures
JSSG-2010	Crew Systems

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE DRAWINGS

200310642	Grinding, High Velocity Oxygen/Fuel Sprayed Tungsten Carbide Coatings

(Copies of this drawing are available at AFLCMC.LZPTP.PublicTORequests@us.af.mil.)

AIR FORCE RESEARCH LABORATORY (AFRL)

AMMTIAC-R000841	Damage Tolerant Design Handbook, MCIC-HB-01

(Copies of this document are available online for DoD users at https://www.dtic.mil and can be located by searching for the accession number given in the left-hand column above.)

AIR FORCE TECHNICAL ORDER

T.O. 1-1-690	General Advanced Composite Repair Processes Manual

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T.O. 34A-3-1	Polymers, Additive Manufacturing, General Procedures and Process Controls for Guidance

(Private industry companies with government contracts may obtain copies of this document by going to http://www.e-publishing.af.mil/ and filling out an AFTO 43 form. The general public may obtain copies by emailing your request to AFLCMC.LZPTP.PublicTORequests@us.af.mil.)

ARMY TECHNICAL MANUAL

TM 1-1500-204-23-11	Aviation Unit Maintenance (AVUM) and Aviation Intermediate Maintenance (AVIM) Manual General Aircraft Maintenance Advanced Composite Material General Maintenance and Practices Volume 11

(Copies of this document are available online at https://armypubs.army.mil.)

DEFENSE TECHNICAL INFORMATION CENTER

ADB096021	DOD/NASA Structural Composites Fabrication Guide, Vol I
ADD436125	DOD/NASA Structural Composites Fabrication Guide, Vol II
ADB080181	DOD/NASA Advanced Composites Design Guide, Vol I
ADB080182	DOD/NASA Advanced Composites Design Guide, Vol II
ADB080183	DOD/NASA Advanced Composites Design Guide, Vol III
ADB080184	DOD/NASA Advanced Composites Design Guide, Vol IV

(Copies of these documents are available online at https://dtic.mil.)

DEPARTMENT OF DEFENSE INSTRUCTION

DoDI 5000.93	Use of Additive Manufacturing in the DoD

(Copies of this document are available online at https://www.acq.osd.mil/se/.)

FEDERAL AVIATION ADMINISTRATION (FAA) ADVISORY CIRCULAR

AC 43-214A	Repairs and Alterations to Composite and Bonded Aircraft Structure

(Copies of this document are available online at: http://www.faa.gov.)

FEDERAL AVIATION REGULATION (FAR)

FAR 25.853	Compartment Interiors Amendment 25-59 Appendix F, Part II, Flammability of Seat Cushions
FAR 25.853	Compartment Interiors Amendment 25-61 Appendix F, Part IV, Test Method to Determine the Heat Release Rate from Cabin Materials Exposed to Radiant Heat

(Copies of these documents are available online at: http://www.faa.gov.)

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NASA TECHNICAL STANDARDS SYSTEM (NTSS)

MSFC-STD-3029	Guidelines for the Selection of Metallic Materials for Stress Corrosion Cracking Resistance in Sodium Chloride Environments
NASA-STD-6030	Additive Manufacturing Requirements for Spaceflight Systems

(Copies of these documents are available online at: https://standards.nasa.gov.)

NAVAIR TECHNICAL PUBLICATIONS

NAVAIR 01-1A-21	Technical Manual Organizational and Intermediate Maintenance General Composite Repair
NAVAIR 01-1A-22	Technical Manual Maintenance Instructions Organizational, Intermediate and Depot Aircraft Radomes and Antenna

(Copies of these documents are available online at: https://mynatec.navair.navy.mil. NATEC has three difference processes to obtain the manuals, depending on if you are Government personnel (military and civilians), contractors with a Government contract, or everyone else. If unable to access this web site or require further assistance, contact (1) NATEC Customer Service: 619.545.1888, (2) NATEC Website Status Hotline: 619.545.1706, or (3) NATEC Customer Service E-mail: nani_customerservice@navy.mil.)

NAWCAD LKE DESIGN DATA REPORT

NAWCADLKE-BL2000-DDR-001	Process Specification for Fused Deposition Modeling of Ultem 9085 CG

(Copies of this document are available at lke-navairdoccom.fct@navy.mil.)

OFFICE OF THE SECRETARY OF DEFENSE MANUFACTURING TECHNOLOGY (OSD MANTECH)

DoD AM Strategy	Department of Defense Additive Manufacturing Strategy

(Copies of this document are available online at https://www.cto.mil/dod-additive-manufacturing-strategy.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS C3.2M/C3.2	Standard Method for Evaluating the Strength of Brazed Joints
AWS C3.3	Recommended Practices for the Design, Manufacture and Examination of Critical Brazed Components

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AWS C3.4M/C3.4	Specification for Torch Brazing
AWS C3.5M/C3.5	Specification for Induction Brazing
AWS C3.6M/C3.6	Specification for Furnace Brazing
AWS C3.7M/C3.7	Specification for Aluminum Brazing
AWS C6.2/C6.2M	Specification for Rotary Friction Welding of Metals
AWS C7.4/C7.4M	Process Specification and Operator Qualification for Laser Beam Welding
AWS D17.1/D17.1M	Specification for Fusion Welding for Aerospace Applications
AWS D17.2/D17.2M	Specification for Resistance Welding for Aerospace Applications
AWS D17.3/D17.3M	Specification for Friction Stir Welding of Aluminum Alloys for Aerospace Applications
AWS D20.1/D20.1M	Specification for Fabrication of Metal Components using Additive Manufacturing
AWS G2.4/G2.4M	Guide for the Fusion Welding of Titanium and Titanium Alloys

(Copies of these documents are available online at https://www.aws.org.)

AMERICAN SOCIETY of MECHANICAL ENGIINEERS (ASME)

ASME B46.1	Surface Texture

(Copies of these documents are available online at https://www.asme.org.)

ASTM INTERNATIONAL

ASTM B265	Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate
ASTM B661	Standard Practice for Heat Treatment of Magnesium Alloys
ASTM D1003	Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics
ASTM D3039/D3039M	Standard Test Method for Tensile Properties of Polymer Matrix Composite Materials
ASTM D4417	Standard Test Method for Field Measurement of Surface Profile of Blast Cleaned Steel
ASTM D4762	Standard Guide for Testing Polymer Matrix Composite Materials
ASTM E23	Standard Test Method for Notched Bar Impact Testing of Metallic Materials
ASTM E162	Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source
ASTM E1417/E1417M	Standard Practice for Liquid Penetrant Testing
ASTM E1444	Standard Practice for Magnetic Particle Testing for Aerospace
ASTM E1570	Standard Practice for Computed Tomographic (CT) Examination
ASTM E1742/E1742M	Standard Practice for Radiographic Examination

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ASTM E1817	Standard Practice for Controlling Quality of Radiological Examination by Using Representation Quality Indicators (RQIs)
ASTM E2375	Standard Practice for Ultrasonic Testing of Wrought Products
ASTM E3375	Standard Practice for Cone Beam Computed Tomographic (CT) Examination
ASTM F733	Standard Practice for Optical Distortion and Deviation of Transparent Parts Using the Double-Exposure Method
ASTM F1165	Standard Test Method for Measurement of Focal Spots of Industrial X-Ray Tubes by Pinhole Imaging
ASTM F3091	Standard Specification for Powder Bed Fusion of Plastic Materials
ASTM F3529	Guide for Additive Manufacturing - Design – Material Extrusion of Polymers
ASTM F3572	Standard Practice for Additive Manufacturing - General Principles - Part Classifications for Additive Manufactured Parts Used in Aviation
ASTM F3606	Standard Guide for Additive Manufacturing - Feedstock Materials - Testing Moisture Content in Powder Feedstock
ISO/ASTM 52900	Additive manufacturing - General principles - Fundamentals and vocabulary
ISO/ASTM 52901	Standard Guide for Additive Manufacturing – General Principles – Requirements for Purchased AM Parts
ISO/ASTM 52903-1	Additive manufacturing - Material extrusion-based additive manufacturing of plastic materials-Part 1: Feedstock materials
ISO/ASTM 52903-2	Additive manufacturing – Material extrusion-based additive manufacturing of plastic materials – Part 2: Process equipment
ISO/ASTM 52936-1	Additive manufacturing of polymers - qualification principles - Part 1: General principles and preparation of test specimens for PBF-LB

(Copies of these documents are available online at https://www.astm.org.)

BATTELLE MEMORIAL INSTITUTE

MMPDS	Metallic Materials Properties Development and Standardization

(Copies of the current edition of MMPDS may be obtained in several forms, as described in www.mmpds.org or by e-mail at bcommpds@battelle.org.)

NATIONAL AEROSPACE STANDARD (NAS)

NAS 410	NAS Certification and Qualification of Nondestructive Test Personnel

(Copies of this document are available online at https://www.aia-aerospace.org/standards.)

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SAE INTERNATIONAL

SAE AIR1388	Service Damage Limits for Aircraft Hydraulic Tubing
SAE AIR1529	Flexure Testing of Hydraulic Tubing Joints and Fittings by Planar Resonant Vibration (Free-Free Beam)
SAE AIR4938	Composite and Bonded Structure Technician/Specialist Training Document
SAE AIR6825	Identification and Assessment of Damage to Composite Aircraft Structures Training Document
SAE AMS2175	Castings, Classification, and Inspection of
SAE AMS2375	Control of Forgings Requiring First Article Approval
SAE AMS2430	Shot Peening
SAE AMS2432	Shot Peening, Computer Monitored
SAE AMS2440	Inspection of Ground, Chromium Plated Steel Parts
SAE AMS2449	Grinding of HVOF Coatings on High Strength Steels
SAE AMS2453	Low Stress Grinding of Steel Parts Heat Treated to 180 ksi or Over, and Low Stress Grinding of Chrome Plating Applied to Steel Parts Heat Treated to 180 ksi or Over
SAE AMS2546	Laser Peening
SAE AMS2590	Rotary Flap Peening of Metal Parts
SAE AMS2631	Ultrasonic Inspection Titanium and Titanium Alloy Bar, Billet and Plate
SAE AMS2632	Ultrasonic Inspection of Thin Materials 0.5 Inch (13mm) and Thinner
SAE AMS2649	Etch Inspection of High Strength Steel Parts
SAE AMS2680	Electron-Beam Welding for Fatigue Critical Applications
SAE AMS2759	Heat Treatment Steel Parts General Requirements
SAE AMS2759/9	Hydrogen Embrittlement Relief (Baking) of Steel Parts
SAE AMS2759/11	Stress Relief of Steel Parts
SAE AMS2761	Heat Treatment of Steel Raw Materials
SAE AMS2768	Heat Treatment of Magnesium Alloy Castings
SAE AMS2770	Heat Treatment of Wrought Aluminum Alloy Parts
SAE AMS2771	Heat Treatment of Aluminum Alloy Castings
SAE AMS2772	Heat Treatment of Aluminum Alloy Raw Materials
SAE AMS2773	Heat Treatment Cast Nickel Alloy and Cobalt Alloy Parts
SAE AMS2774	Heat Treatment Wrought Nickel Alloy & Cobalt Alloy Parts
SAE AMS2801	Heat Treatment of Titanium Alloy Parts
SAE AMS3100	Adhesion Promoter for Polysulfide Sealing Compounds
SAE AMS3216	Fluorocarbon (FKM) Rubber High-Temperature - Fluid Resistant Low Compression Set 70 to 80
SAE AMS3218	Fluorocarbon (FKM) Rubber High-Temperature-Fluid Resistant Low Compression Set 85 to 95
SAE AMS3276	Sealing Compound, Integral Fuel Tanks and General Purpose, Intermittent Use to 360 °F (182 °C)
SAE AMS3277	Sealing Compound, Polythioether Rubber Fast Curing for Integral Fuel Tanks and General Purpose, Intermittent Use to 360 °F (182 °C)

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SAE AMS3278	Sealing and Coating Compound, Polyurethane (PUR) Fuel Resistant High Tensile Strength/Elongation for Integral Fuel Tanks/Fuel Cavities/General Purpose
SAE AMS3279	Sealing Compound, Sprayable for Integral Fuel Tanks and Fuel Cell Cavities, For Intermittent Use to 350 °F (177 °C) (Stabilized Type)
SAE AMS3281	Sealing Compound, Polysulfide (T) Synthetic Rubber for Integral Fuel Tank and Fuel Cell Cavities Low Density for Intermittent Use to 360 °F (182 °C)
SAE AMS3283	Sealing Compound, Polysulfide Non-Curing, Groove Injection Temperature and Fuel Resistant
SAE AMS3374	Sealing Compound, One Part Silicone, Aircraft Firewall
SAE AMS3376	Sealing Compound, Non-Curing, Fluorosilicone Groove Injection Temperature and Fuel Resistant
SAE AMS3381	Sealing Compound, Flame Resistant, Non-Silicone
SAE AMS3384	Rubber, Fluorocarbon Elastomer (FKM) 70 to 80 Hardness, Low Temperature Sealing Tg -22 °F (-30 °C) For Elastomeric Shapes or Parts in Gas Turbine Engine Oil, Fuel and Hydraulic Systems
SAE AMS3856	Cloth, Upholstery, Flame Resistant, Novoloid/Aramid, Waffle Type Weave
SAE AMS4900	Titanium Sheet, Strip, and Plate Commercially Pure Annealed, 55 ksi (379 MPa) Yield Strength - UNS R50550
SAE AMS4901	Titanium Sheet, Strip, and Plate Commercially Pure Annealed, 70.0 ksi (485 MPa) - UNS R50700
SAE AMS4902	Titanium Sheet, Strip, and Plate Commercially-Pure Annealed 40.0 ksi (276 MPa) Yield Strength - UNS R50400
SAE AMS4903	Titanium Alloy Sheet, Strip, and Plate 6Al - 4V Solution Heat Treated - UNS R56400
SAE AMS4904	Titanium Alloy Sheet, Strip, and Plate 6Al - 4V Solution Heat Treated and Aged - UNS R56400
SAE AMS4907	Titanium Alloy, Sheet, Strip, and Plate 6.0Al - 4.0V, Extra Low Interstitial Annealed - UNS R56401
SAE AMS4909	Titanium Alloy, Sheet, Strip, and Plate 5Al - 2.5Sn, Extra Low Interstitial Annealed - UNS R54521
SAE AMS4910	Titanium Alloy, Sheet, Strip, and Plate 5Al - 2.5Sn Annealed - UNS R54520
SAE AMS4911	Titanium Alloy, Sheet, Strip, and Plate 6Al - 4V Annealed - UNS R56400
SAE AMS4915	Titanium Alloy Sheet, Strip, and Plate 8Al - 1V - 1Mo Single Annealed - UNS R54810
SAE AMS4916	Titanium Alloy Sheet, Strip, and Plate 8Al - 1Mo - 1V Duplex Annealed - UNS R54810
SAE AMS4917	Titanium Alloy Sheet, Strip, and Plate 13.5V - 11Cr - 3.0Al Solution Heat Treated - UNS R58010
SAE AMS4918	Titanium Alloy, Sheet, Strip, and Plate 6AI - 6V - 2Sn Annealed - UNS R56620
SAE AMS4919	Titanium Alloy Sheet, Strip, and Plate 6AI - 2Sn - 4Zr - 2Mo - 0.08Si Duplex Annealed - UNS R54620

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SAE AMS4921	Titanium Bars, Wire, Forgings, and Rings Commercially Pure 70 ksi (483 MPa) Yield Strength - UNS R50700
SAE AMS4939	Titanium Alloy Sheet, Strip, and Plate 3Al - 8V - 6Cr - 4Mo -4Zr Solution Heat Treated - UNS R58640
SAE AMS4940	Titanium Sheet, Strip, and Plate Commercially Pure Annealed, 25.0 ksi (172 MPa) Yield Strength - UNS R50250
SAE AMS4941	Titanium Tubing, Welded, 40 ksi (276 MPa)
SAE AMS4942	Titanium Tubing, Seamless Annealed, 40,000 psi, Yield
SAE AMS4943	Titanium Alloy Tubing, Seamless 3.0Al - 2.5 V, Annealed
SAE AMS4944	Titanium Alloy Tubing, Seamless Hydraulic, 3.0Al - 2.5 V, Cold Worked, Stress Relieved
SAE AMS4945	Titanium Alloy Tubing, Seamless Hydraulic, 3.0Al - 2.5 V, Texture Controlled, 105,000 psi (724 MPa) Yield Strength
SAE AMS4970	Titanium Alloy Bars, Wire, and Forgings 7Al - 4Mo Solution and Precipitation Heat Treated - UNS R56740
SAE AMS4988	Titanium Alloy Sheet, Strip, and Plate 6Al - 6V - 2Sn Solution Heat Treated - UNS R56620
SAE AMS4989	Titanium Alloy Sheet, Strip, and Plate 3Al - 2.5V Annealed - UNS R56320
SAE AMS4990	Titanium Alloy Sheet, Strip, and Plate 6Al - 6V - 2Sn Solution Heat Treated and Aged - UNS R56620
SAE AMS6900	Titanium Alloy Bars, Forgings and Forging Stock 5Al - 2.5Sn Annealed - UNS R54520
SAE AMS6901	Titanium Alloy Bars, Forgings, and Forging Stock 5Al - 2.5Sn, Extra Low Interstitial Annealed - UNS R54521
SAE AMS6905	Titanium Alloy Bars, Forgings and Forging Stock 6.0Al - 2.0Sn - 4.0Zr - 2.0Mo Duplex Annealed - UNS R54620
SAE AMS6906	Titanium Alloy Bars, Forgings, and Forging Stock 6.0Al -2.0Sn - 4.0Zr - 6.0Mo Solution Heat Treated and Aged - UNS R56260
SAE AMS6907	Titanium Alloy Bars, Forgings and Forging Stock 6.0Al - 2.0Sn - 4.0Zr - 6.0Mo Duplex Annealed - UNS R56260
SAE AMS6910	Titanium Alloy Bars, Forgings and Forging Stock 8Al - 1Mo 1V Duplex Annealed - UNS R54810
SAE AMS6915	Titanium Alloy Bars, Forgings and Forging Stock 7.0Al - 4.0Mo Annealed - UNS R56740
SAE AMS6920	Titanium Alloy Bars, Forgings and Forging Stock 3Al - 8V - 6Cr - 4Mo - 4Zr Solution Heat Treated - UNS R58640
SAE AMS6921	Titanium Alloy Bars, Forgings and Forging Stock 3Al - 8V - 6Cr - 4Mo - 4Zr Solution Heat Treated and Aged - UNS R58640
SAE AMS6925	Titanium Alloy Bars, Forgings and Forging Stock 13V - 11Cr - 3Al Solution Heat Treated - UNS R58010
SAE AMS6926	Titanium Alloy Bars, Forgings and Forging Stock 13V - 11Cr - 3Al Solution Heat Treated and Aged - UNS R58010
SAE AMS6930	Titanium Alloy Bars, Forgings and Forging Stock 6.0Al - 4.0V Solution Heat Treated and Aged - UNS R56400

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SAE AMS6931	Titanium Alloy Bars, Forgings and Forging Stock 6.0Al - 4.0V Annealed - UNS R56400
SAE AMS6932	Titanium Alloy Bars, Forgings and Forging Stock 6.0Al – 4.0V Extra Low Interstitial Annealed - UNS R56401
SAE AMS6935	Titanium Alloy Bars, Forgings and Forging Stock 6.0Al 6.0V – 2.0Sn Solution Heat Treated and Aged - UNS R56620
SAE AMS6936	Titanium Alloy Bars, Forgings and Forging Stock 6Al - 6V 2Sn Annealed - UNS R56620
SAE AMS6940	Titanium Alloy Bars, Forgings and Forging Stock 3.0Al - 2.5V Annealed - UNS R56320
SAE AMS7101	Fused Filament Fabrication, Material for
SAE AMS7100/1	Fused Filament Fabrication Process, Stratasys Fortus 900mc plus with Type1, Class 1, Form 1, Grade 0, Natural Color Material for
SAE AMS7259	Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant, Low Compression Set / 85 to 95 Hardness, For Seals in Fuel Systems and Specific Engine Oil Systems
SAE AMS7276	Rubber: Fluorocarbon (FKM), High-Temperature-Fluid Resistant, Low Compression Set, For Seals in Fuel Systems and Specific Engine Oil Systems
SAE AMS7287	Fluorocarbon Elastomer (FKM) High Temperature/HTS Oil Resistant/Fuel Resistant, Low Compression Set / 70 to 80 Hardness, Low Temperature Tg -22 °F (-30 °C) For Seals in Oil / Fuel / Specific Hydraulic Systems
SAE AMS7379	Rubber: Fluorocarbon Elastomer (FKM), 70-80 Hardness, Low Temperature Sealing Tg -40°F (-40°C) For Elastomeric Seals in Aircraft Engine Oil, Fuel and Hydraulic Systems
SAE AMS7410	Rubber: Fluorocarbon Elastomer (FKM) Aircraft Engine Oil, Fuel and Hydraulic Fluid Resistant 70-80 Shore Type A Hardness, Low Temperature Sealing Tg -47°F (-43.9°C) For Elastomeric Seals in Aircraft Engine, Fuel and Hydraulic Systems
SAE AMS-C-7438	Core Material, Aluminum, for Sandwich Construction
SAE AMS-C-8073	Core Material, Plastic Honeycomb, Laminated Glass Fabric Base, for Aircraft Structural and Electronic Applications
SAE AMS-C-27725	Coating, Corrosion Preventative, for Aircraft Integral Fuel Tanks for Use to 250 °F (121 °C)
SAE AMS-H-81200	Heat Treatment of Titanium and Titanium Alloys
SAE AMS-P-5315	Acrylonitrile - Butadiene - (NBR) Rubber for Fuel-Resistant Seals 60 to 70
SAE AMS-P-5510	O-Ring, Preformed, Straight Thread Tube Fitting Boss, Type I Hydraulic (-65° to 160°F)
SAE AMS-P-83310	Plastic Sheet, Polycarbonate, Transparent
SAE AMS-P-83461	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 Degrees F (135 Degrees C)
SAE AMS-R-7362	Nitrile Rubber, Synthetic, Solid, Sheet, Strip and Fabricated Parts, Synthetic Oil Resistant
SAE AMS-R-83285	Rubber, Ethylene-Propylene, General Purpose -

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SAE AMS-S-8802	Sealing Compound, Fuel Resistant, Integral Fuel Tanks and Fuel Cell Cavities
SAE AMS-S-83318	Sealing Compound, Polysulfide Type, Low Temperature Curing, Quick Repair, Integral Fuel Tanks and Fuel Cell Cavities
SAE AMS-STD-2154	Inspection, Ultrasonic, Wrought Metals, Process for
SAE ARP4462	Barkhausen Noise Inspection for Detecting Grinding Burns in High Strength Steel Parts
SAE ARP7488	Peening Design and Process Control Guidelines
SAE AS291	Surface Finish (AA)
SAE AS568	Aerospace Size Standard for O-Rings
SAE AS603	Impulse Testing of Hydraulic Hose, Tubing, and Fitting Assemblies
SAE AS1933	Age Controls for Hose Containing Age-Sensitive Elastomeric Material
SAE AS3208	Packing, Preformed - AMS 7276 - Seal
SAE AS3209	Packing, Preformed - AMS 7276, 'O' Ring
SAE AS3581	Packing, Preformed - O-Ring Seal AMS 7259
SAE AS4265	Impulse Testing of Hydraulic Tubing & Fittings, S-N Curve
SAE AS4467	Standardization, Hydraulic Tubing Walls
SAE AS5316	Storage of Elastomer Seals and Seal Assemblies Which Include an Elastomer Element Prior to Hardware Assembly
SAE AS6279	Standard Practice for Production, Distribution and Procurement of Metal Stock
SAE AS6870	Non-Destructive Program Requirements for Aerospace Systems
SAE AS7045	Residual Stress Measurement and Classification, Metallic Structural Alloy Products and Finished Parts
SAE AS12500	Corrosion Prevention and Deterioration Control in Electronic Components and Assemblies.
SAE AS28775	Packing, Preformed - MS28775 O-Ring
SAE AS28778	O-Ring, Straight Thread Tube Fitting Boss, Molded From AMS-P-5510 Rubber
SAE AS29512	Hydrocarbon Fuel Resistant, Tube Fitting, O-Ring, Molded From AMS-P-5315 Rubber
SAE AS29513	Packing, Preformed, Hydrocarbon Fuel Resistant O-Ring
SAE AS29561	O-Ring, Synthetic Lubricant Resistant, Molded From AMS-R-7362 Rubber
SAE AS50881	Wiring, Aerospace Vehicle
SAE AS81550	Insulating Compound, Electrical, Embedding, Reversion Resistant Silicone
SAE AS83461/1	M83461 O-Ring Molded From AMS-P-83461 Rubber
SAE AS83461/2	Straight Thread Tube Fitting Boss O-Ring Molded From AMS-P-83461 Rubber
SAE AS85421	Fittings, Tube, Fluid Systems, Separable, Beam Seal 3000/4000 psi, General Specification for
SAE AS85720	Fittings, Tube, Fluid Systems, Separable, High Pressure, Dynamic Beam Seal, General Specification for

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SAE CMH-17-1	Polymer Matrix Composites, Guide for Characterization of Structural Materials
SAE CMH-17-2	Polymer Matrix Composites, Materials Properties
SAE CMH-17-3	Polymer Matrix Composites, Materials Usage, Design and Analysis
SAE CMH-17-4	Metal Matrix Composites
SAE CMH-17-5	Ceramic Matrix Composites
SAE CMH-17-6	Structural Sandwich Composites
SAE MA2003	Rotary Flexure Testing of Hydraulic Tubing Joints and Fittings
SAE MA2008	Radii-Tube Bend, Metric

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Additive manufacturing (AM) processes. Additive manufacturing (AM) processes to create parts from three-dimensional model data, usually by adding layer on layer, also known as three-dimensional printing (e.g. ISO/ASTM 52900). These processes include, but are not limited to, processes such as powder bed fusion, fused filament fabrication, stereo-lithography additive, binder jetting and directed energy deposition used to produce components or portions of components. For the purposes of this standard, processes for repair or geometric restoration that generally do not rely on three-dimensional model data, such as cold spray processes, multi-pass robotic welding, etc. are included due to similarity of qualification approach. Additive manufacturing does not include coating processes such as those covered in MIL-DTL-5002.

3.2 Aerospace weapons systems. Applicable systems include fixed wing aircraft, unmanned aerial vehicles, space vehicles, rotorcraft, missile systems, air delivered guided munitions, armament systems and all related ground support equipment.

3.3 Cognizant engineering authority (CEA). The cognizant engineering authority is that entity designated by the procuring activity that is responsible for design approval; reviewing and approving the technical requirements contained within this standard; and approving any deviations from them.

3.4 Composites. Composites are material systems made up of a reinforcing material and a matrix. Composite materials are divided into three broad categories: organic matrix composites, ceramic matrix composites, and metal matrix composites. Organic matrix composites consist of organic resins reinforced with fibers, i.e., fiberglass, carbon (graphite), aromatic polyamide (aramid), etc. The fiber reinforcement takes the form of continuous unidirectional filaments (tape), woven fabric (cloth), chopped fibers, etc. Higher performing organic composites contain organic resins reinforced with high strength, high stiffness fibers such as boron or carbon (graphite). Ceramic matrix composites are fiber, whisker, or particulate reinforced ceramics. Metal matrix composites are fiber, whisker, or particulate reinforced metals.

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3.5 Corrosion. Corrosion is the deterioration of a material or its properties due to a reaction of that material with its environment. Corrosion includes degradation of non-metallic materials. Some non-traditional examples include rotting of wood, degradation of concrete (carbonation, alkali-silica reaction phenomena), and degradation of composite materials due to reaction with the environment.

3.6 Criticality/consequence of failure. Criticality or consequence of failure for aerospace flight parts is defined by each agency depending on the outcome of the failure and is broken out into three or four categories depending on the government (e.g., MIL-STD-1530, NASA-STD-6030) or industry (e.g., ASTM F3572, AWS D20.1/D20.1M) standards approved for use by their CEA. For this document, the following categories are used to describe criticality. These should be translated to the appropriate specification based on the requirements of the CEA. Non-flight parts should be managed by local policy.

a. Critical (NASA-STD-6030 Class A). Failure of aircraft part may lead to loss of aircraft, loss of stores, inability to release stores, abortion of mission, or may endanger personnel. Terms such as safety-of-flight, critical safety item, fracture critical, and mission critical are associated with this classification.

b. Semi-critical (NASA-STD-6030 Class B). Failure of aircraft part may reduce system efficiency but does not lead to loss of aircraft or endanger personnel. Terms such as durability critical and maintenance critical are associated with this classification.

c. Non-critical (NASA-STD-6030 Class C). Failure of component or repair will not significantly affect other system elements or pose a danger to personnel. Terms such as durability non-critical and normal controls are associated with this classification.

3.7 Damage. Damage is the presence of any crack, flaw, corrosion, disbond, delamination, or other feature that degrades, or has the potential to degrade the performance of the affected part.

3.8 Defect. A defect is any discontinuity or discrepancy that negatively impacts the service performance of the part.

3.9 Durability. Durability is the ability of the structure and components to resist cracking, corrosion, disbanding, thermal or environmental degradation, delamination, wear, and the effects of foreign object damage for a prescribed period of time.

3.10 Exterior surfaces. All surfaces normally exposed to an external environment are considered exterior surfaces. All interior surfaces that may also become wetted with water or corrosive fluid are considered exterior surfaces. These surfaces include, but are not limited to, wheels and landing gear, wheel wells, and their fairings, control surfaces, wing-fold areas, battery compartments, and bilge areas on aircraft with latrines.

3.11 In-process correction. An in-process correction is an unplanned action performed on a nonconforming part at any point prior to final inspection in order to meet all requirements and make it acceptable for the intended use, pending review and approval by the CEA. Examples include, but are not limited to bending, blending, sanding, peening, grinding, machining, composite patching, welding, or brazing for the purposes of defect removal or part restoration to drawing allowances. For additive manufacturing processes an example would be a restart after an un-planned build interruption but does not apply to changes to previously qualified/approved

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AM process parameters.

3.12 Lubricants. Lubricants include lubricating oils, greases, solid film lubricants, and anti-seize compounds used to reduce friction and wear.

3.13 Material specification. The material specification should contain all material requirements, including process controls, acceptance testing (typically mechanical, chemical, or thermal analysis) and acceptance criteria for all tests.

3.14 Non-destructive inspection (NDI). Non-destructive inspection is an inspection process or technique designed to reveal the damage at or beneath the external surface of a part or material without adversely affecting the material or part being inspected. NDI generally refers to inspections that are conducted using equipment that is not part of or permanently affixed to the part being inspected.

3.15 Organic resins for composites. The organic resin (binder, resin, and matrix are interchangeable terms) of organic matrix composites can be a thermoset or a thermoplastic polymer. A thermoset composite is processed to a product form by a chemical reaction known as cure. The curing reaction can be facilitated by heat and pressure, as in an autoclave cure, or by other means such as radio frequency, or radiation exposure. Typically, the cure temperature can be room temperature, 121 °C - 176 °C (250 °F - 350 °F) for epoxies, approximately 232 ºC (450 ºF) for bismaleimides, and 288 ºC - 371 ºC (550 ºF - 700 ºF) for polyimides, etc. A thermoplastic composite is processed to a product form by operations such as deformation forming, injection molding or consolidation molding conducted at elevated temperatures where viscous flow of the matrix is possible.

3.16 Process control factor plan. The process control factor plan should include a list of process control factors for the specified components based on their criticality and the verification of process capability, qualification, monitoring, and control of variation over time.

3.17 Process control factors. Process control factors are those where the output cannot be or, is difficult to validate after component fabrication and can have a significant impact on product integrity. Because these processes cannot be validated after completion, additional oversight is needed to ensure integrity of the process. Examples include heat treatments, welding and bonding, composite fabrication, additive manufacturing, shot peening, nondestructive inspection, sealant application, and organic and inorganic finish application.

3.18 Process control reference distribution (PCRD). A PCRD is a statistical model representing a material property used for monitoring process control. A PCRD is developed for material produced to an approved qualified material process (QMP) and tested with a consistent specimen geometry, size, orientation, and test procedure. A PCRD is used to define the acceptance criteria applied to selected witness tests for an AM build cycle; the data used to develop the statistical PCRD model are typically the result of nominal witness tests and other data generated with the same consistent specimen geometry and test procedure.

3.19 Prolongation. Prolongation is an extra portion of material added in a mutually agreeable location of a forging, casting or composite part to permit removal and subsequent testing without destroying the part.

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3.20 Restricted materials. These are materials that require approval by the CEA prior to use based on regulations, known deficiencies, or deleterious effects from their use in aerospace weapons systems.

3.21 Steel. For purposes of this standard, steel is an alloy of iron and carbon with or without other alloying elements. 5.1.7 identifies common steel alloys used in aerospace weapons systems grouped into various broad groups based on microstructures produced by various heat treatments.

3.22 Stress corrosion cracking (SCC). Cracking induced from the combined influences of sustained tensile stress and a corrosive environment on a susceptible material. SCC cracks grow at unpredictable rates and pose a flight safety risk.

3.23 Supportability. Supportability is the degree to which acceptable quality and cost-effective preventive methods, in-service repair methods, or replacements are either available or can be developed in a timely manner to mitigate the deterioration of materials and structures fabricated using the selected manufacturing processes and joining methods.

3.24 Trim. Trim is the excess metal or flash removed during the forging or casting process. The trimming operation takes place in tools produced to the peripheral shape of the component with the component being pushed through the female impression by the identically shaped male punch. Trimming may be carried out at room temperature or elevated temperature.

3.25 Wear. Wear is the damaging, gradual removal, or deformation of a material at solid surfaces. Causes of wear can be mechanical (e.g., erosion, abrasion, fretting) or chemical (e.g., corrosion).

4. GENERAL REQUIREMENTS

4.1 Materials and processes selection. The selection of materials and processes shall be determined by design studies for each system. The studies shall include all the relevant parameters specified by the contract, such as operational environments, performance, manufacturing capabilities, safety of flight structure, life cycle costs, and reliability and maintainability requirements. The studies shall determine the acceptable initial flaw types and size, defects, and tolerances associated with the manufacturing processes, fabrication, and assembly. The initial manufactured quality of design shall meet the service requirements specified in the contract including, but not limited to fatigue, rigidity, strength, durability, and damage tolerance. Trade studies for material selection shall include the mechanical properties relevant to the design application and identified documented sources approved by the CEA [e.g., the Metallic Materials Properties Development and Standardization (MMPDS) Handbook or the composites materials handbook (CMH-17)]. All materials and processes selected for use in aerospace weapons systems shall have material specifications and process specifications (i.e., Government, Non-Government, or Industry) reviewed and approved by the CEA as defined in 4.4 in order to ensure consistent quality, conformability, and repeatability. The protective surface treatments and finishing systems used on weapons systems parts (including spares) and assemblies (such as fuselages, wings, cowlings, empennage, and rotor blades) are outlined in 4.5. In-service performance data of materials and processes in similar applications shall be considered when data is available. When new or alternative materials or processes with limited in-service performance data are proposed, their use shall require approval by the CEA.

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4.2 Restricted structural alloys. The materials listed in table I are commonly restricted from use in structural applications in new design on Department of Defense (DoD) weapons systems. Any use of these materials in new design requires specific approval from the CEA. Aerospace weapons systems parts that are currently in-service (i.e., legacy aircraft using traditional metals and alloys, heat treatments and product forms) are not restricted or prohibited for procurement of spares or replacement parts where there are no new drawing changes or modifications and where in-service data shows acceptable performance and low risk for continued use. SCC and other corrosion related thickness limitations shall be in accordance with MIL-STD-1568.

4.3 Material properties/fracture mechanics. For aerospace weapons systems, static, dynamic, chemical, and physical properties, as well as corrosion resistance and producibility shall be evaluated for the specific design application. Because of the extreme variation in environmental conditions where weapons systems may be used or stored, it is imperative to acquire high confidence data that has been reviewed (along with the analysis techniques applied) and accepted by the CEA. At a minimum, static properties [e.g., tensile strength (TS), yield strength (YS), elongation (EL) and elastic modulus (Mod)], dynamic properties (e.g., toughness, fatigue crack growth rates and impact resistance), physical properties, SCC thresholds, corrosion resistance as required for the application, thermal stability, and producibility (machining, welding, and forming) shall be evaluated. Fracture mechanics analysis shall be in accordance with MIL-STD-1530, MIL-STD-1798 and MIL-STD-3024. MCIC-HB-01 shall be used to establish material properties when conducting fracture mechanics analysis.

4.4 Material and process specifications. The materials and processes submitted in the technical proposal for the aerospace weapons system shall be selected from CEA approved Government (DoD, Fed, etc.), non-Government (AMS, AWS, ASTM, etc.) and industry proprietary specifications and standards. At a minimum, the technical proposals shall contain a description of the operating environment and loading conditions and a listing of all materials and processes along with their selection rationale. For cases where an approved government or industry specification does not exist for a proposed material or process, a contractor prepared specification, containing rationale and tests to be performed, shall be prepared in accordance with DI-MFFP-82119 for approval by the CEA.

4.4.1 Material specification. A material specification shall be established in accordance with DI-MFFP-82119 (see 6.3). The specification at a minimum shall identify in detail.

a. the types of chemical, mechanical, and other tests to be performed.
b. the types and numbers of test specimens to be employed.
c. the range of test results that are needed for the incoming material to be acceptable for manufacturing.
d. the conformance inspections (sampling and frequency) for individual purchase lots of incoming material after that material has been approved.
e. shelf life and storage requirements for organic materials.

4.4.2 Process specification. A process specification shall be established in accordance with DI-MFFP-82119 (see 6.3). The process specifications shall provide for and detail each process that is essential to fabricate or procure a product or material, particularly the processing of the raw material or preform into an assembly component conforming to engineering design requirements. Process control shall include a verification method and frequency of performance, preferably companion (or traveler) coupons that accompany all production parts during the manufacturing

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process cycle. These coupons shall be tested to verify that the manufacturing cycle has produced parts within an acceptable range.

4.5 Corrosion/finishing requirements. Corrosion prevention and control shall be in accordance with MIL-STD-1568, MIL-STD-889 and MIL-STD-810 (Method 509.8). Cleaning and inorganic surface treatments shall be in accordance with MIL-DTL-5002. Organic finishes, coatings, and sealants shall be in accordance with MIL-STD-7179.

4.6 Material property design allowables. The material property design allowables shall be developed according to a material characterization and design allowable substantiation plan which defines the requirements for characterizing the proposed materials of construction and their respective design allowables. Design allowables based on static and fatigue test data shall be substantiated and analyzed in accordance with procedures used for corresponding data in the appropriate handbook (e.g., MMPDS or CMH-17). The substantiation plan shall be approved by the CEA. In addition to static properties, all relevant critical material properties such as fatigue, creep, impact damage resistance, corrosion resistance, SCC resistance, etc. shall be assessed in the design allowables. Mechanical property design allowables shall be derived in accordance with MMPDS or CMH-17-1 statistical procedures from a statistically significant amount of test data subject to review and approval by the CEA. The validity of test data and the derivation of design allowables from the test data shall be documented and approved by the CEA. The preparation of test specimens to verify properties shall be in accordance with the same material and manufacturing process specifications proposed for use in the systems production. Quality control of material properties and processes during manufacture shall be in accordance with the CEA approved detailed testing, inspection, and process controls in accordance with the governing specifications.

4.7 Joining and fastening. Fastening methods (bonding, bolting, stitching, welding, brazing, soldering, etc.) for the chosen materials and processes shall be reviewed and approved by the CEA. The effect of joining and fastening techniques on the net material property design allowables shall be evaluated, including sensitivities to material and process variations such as hole quality, bondline thickness, adhesive bonding surface preparation, stitching quality, thermal variations from joining processes, etc. Joint system design allowables, test methods, and data reduction methods for determining those allowables are described in MMPDS and CMH-17. Hole tolerances and fastener pull-through resistance along with corrosion resistance in accordance with MIL-STD-1568 and MIL-STD-889 shall be evaluated during design development. Attaching parts such as nuts, bushings, spacers, washers, screws, self-tapping screws, sleeves, self-locking nuts, speed nuts, clamps, and bolts shall be installed in accordance with MIL-STD-7179 and MIL-STD-889. Protective finishing requirements for fastener installation shall be in accordance with MIL-STD-1568.

4.8 Additive manufacturing. Additive manufacturing (AM) processes shall be used in accordance with the DoD AM Strategy and DoDI 5000.93. Material and process requirements for AM are detailed in 5.4.

4.9 Electrical/electromagnetic behavior. The electrical/electromagnetic performance behavior of the material including its effect on galvanic compatibility in accordance with MIL-STD-889 shall be considered in the structural design. Of particular concern are shielding effectiveness, joint design/signature control (conductivity, corrosion, maintainability, etc.), lightning strike protection, fuel tank design (spark-free fuel tanks, etc.), power systems

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grounding, high and low frequency antenna performance, analytical techniques, and combined space environmental effects.

4.10 Repairability. The materials and processes selected for design of aerospace weapons systems shall take into consideration the ability to be repaired and maintained utilizing existing materials, technology, and capabilities within the Military Departments. This applies to both depot and field operations. When existing capabilities cannot achieve this support, the design shall include an analysis of all new repair or maintenance technologies required.

4.11 Thermal expansion compatibility. The effects of induced stresses resulting from thermal expansion differences (i.e., the mating of materials with different thermal expansion coefficients) shall be evaluated during the design and manufacturing process.

4.12 Supportability. Materials and processes for aerospace weapons systems shall be supportable and correlated with mechanical property design allowable data. Supportability means that thermal, environmental, and mechanical deterioration of materials have been identified and that acceptable quality and cost-effective preventive methods and in-service repair methods are either available or can be developed in a timely manner to mitigate the deterioration of materials and structures fabricated using the selected manufacturing processes and joining methods.

4.13 Environmental compliance. Materials and processes for aerospace weapons systems shall be in compliance with federal, state, and local environmental rules and regulations.

4.14 Electronic components and assemblies. Materials and processes used for aerospace weapons systems electronic components and assemblies shall be in accordance with SAE AS12500.

5. DETAILED REQUIREMENTS

5.1 Metals and alloys.

5.1.1 Material selection. The selection of metals and alloys shall comply with the detailed requirements of section 4 of this standard (material and process specifications, corrosion and finishing, material allowables, restricted materials, etc.). Acceptable static property material allowables for AMS specification compliant metals and alloys are found in MMPDS. To ensure maximum corrosion resistance the selection of metals and alloys shall meet the requirements of MIL-STD-1568. In accordance with SAE AS6279, intermediate product forms (e.g., billet, reforging stock, rerolling stock, ingots, etc.) shall not be used to directly manufacture parts or components. Additional information and guidance for selection of metals and alloys for resistance to SCC in sodium chloride environments is provided in MSFC-STD-3029. When coupling of dissimilar metals or alloys in environments where an electrolyte cannot be avoided and galvanic corrosion is predicted, the requirements of 4.5 shall apply. Metals or alloys and heat treatment conditions that have a ductile to brittle transition shall not be used in applications where the service environment is less than 50 ºF above the ductile to brittle transition temperature (DBTT) determined by impact testing or reference temperature (T0) determined by fracture mechanics or other methods approved by the CEA.

5.1.2 Incompatible metal to metal contact. Galvanic corrosion protection between metals

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shall be in accordance with MIL-STD-889 and MIL-STD-1568. In addition to galvanic corrosion, there are other chemical reactions between metals and alloys that may cause chemical and metallurgical reactions that result in deterioration, amalgamation, embrittlement, and premature failure. These reactions may occur at sub-ambient, ambient, or elevated temperatures and do not necessarily require an electrolyte (e.g., amalgamation between the metals). There are combinations of metals and alloys in aerospace where contact between the two can result in liquid metal embrittlement (LME) or solid metal embrittlement (SME). The following metal to metal contact between parts, parts and fixtures, and parts and tools are prohibited unless specifically approved by the CEA:

a. Cadmium in contact with titanium and titanium alloys at any temperature (cadmium tools, fixtures, etc. shall not be used on titanium parts, components, or assemblies). Cadmium can cause both SME and LME of titanium alloys.

b. Cadmium plating on parts where service temperatures exceed 450 ºF. Cadmium on steels at temperatures above 450 ºF can cause LME. See maximum temperature limits for coating and platings specified in MIL-DTL-5002.

c. Silver in contact with titanium (can cause SME and LME) and silver in contact with nickel-based alloys at temperatures of 1000 ºF or higher.

d. Mercury or gallium (from thermometers, electronics) in contact with aluminum (produces amalgamation or liquid metal embrittlement of the aluminum if the protective oxide is disrupted).

5.1.3 Residual stress relief. Undesirable residual stresses (tension or compression) are the stresses in a material or component that are present with no external force. Residual stresses typically arise during the thermomechanical processing of raw materials and can be worsened by part manufacturing processes. Undesirable residual stresses may also occur in service due to unanticipated loading during assembly or operation. Residual stresses in aluminum, titanium, nickel, and steel, shall be minimized using CEA approved thermal or mechanical (e.g., stress relief by stretching of aluminum alloys or compression) stress relieving treatments. These stress relieving processes may be part of the heat treatment process (e.g., artificial aging of precipitation hardening alloys, tempering of steels, etc.). The primary operational benefit of stress relief is more consistent and reliable performance of components in service. Other benefits of stress relief include better dimensional stability during machining or heat treatment, reduced propensity for cracking during plating and minimizing the occurrence of hydrogen embrittlement or SCC. Aluminum, titanium, nickel, and steel alloy parts produced without detrimental residual stresses may be used without stress relief treatments with approval by the CEA. Residual stress measurements, if required, shall be in accordance with SAE AS7045 or other methods approved by the CEA.

5.1.4 Engineered residual stresses. Engineered residual stresses are beneficial residual stresses in components imparted through controlled processes such as peening, cold hole expansion, burnishing, or other cold working processes. Engineered residual stresses have the benefits of improving fatigue performance, corrosion resistance, SCC resistance, resistance to hydrogen absorption, and counteracting harmful residual stresses from coatings or platings. When used, shot peening shall be in accordance with SAE AMS2430 or SAE AMS2432, rotary flap peening shall be in accordance with SAE AMS2590, and laser shock peening shall be in

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accordance with SAE AMS2546. The optimum peening parameters shall be defined by the CEA and recommended practices are found in accordance with SAE ARP7488. Peening of holes, fillets, threads, runout areas, etc. in steel parts shall be in accordance with SAE AMS2430. The process parameters for other engineered residual stresses processes such as cold hole expansion or low plasticity burnishing are dependent on the component and shall be defined by the CEA. Unless otherwise authorized by the CEA, all steel parts above 180 ksi that are used in fatigue or fracture critical applications shall be shot peened after final machining and before application of surface finishes.

5.1.5 Cleaning. During manufacture or use, raw material and parts shall be cleaned to remove organic and inorganic surface contamination in accordance with MIL-DTL-5002 prior to subsequent processing and shall be clean and visually free of contaminants such as dirt, metal residues, lubricants, and solvent residues.

5.1.6 Aluminum alloys.

5.1.6.1 Heat treatment. Heat treatment of aluminum alloy raw materials, wrought parts, and castings shall be in accordance with SAE AMS2772, SAE AMS2770, SAE AMS2771 respectively. The process control provisions of these specifications shall be used for heat treatment of raw material, wrought parts, or castings made from aluminum alloys not specifically covered in these specifications, provided the temperatures, temperature uniformity tolerances, times, and quenchants are specified and approved by the CEA. If tensile or shear testing is specified by the CEA, test specimens shall represent the maximum thickness of each lot of parts in each heat treat load. CEAs that provide approval for straightening after aging should also consider requiring additional inspections or surveillance to ensure the final part properties are not adversely affected. Special considerations that deviate from these specifications, such as quench medium, delay in final aging, or forming prior to aging shall be approved by the CEA.

5.1.6.2 Straightening during heat treatment. The design of the heat treatment of aluminum alloys (heating equipment, racks, fixtures, quenchant selection, etc.) shall be optimized to minimize warping and distortion during heat treatment of aluminum alloys. For wrought aluminum parts heat treated to SAE AMS2770, raw materials heat treated to SAE AMS2772, straightening of parts (in the final artificially aged tempers or naturally aged to the final hardness), or conductivity minimums are prohibited unless specifically authorized by the CEA. Stretching or compressing for residual stress treatment before aging is permissible if required by the temper designation. Straightening of aluminum castings heat treated in accordance with SAE AMS2771 is prohibited unless approved by the CEA. No impact bending, reverse bending or localized bending is allowed.

5.1.7 Steels. The term “steel” in this section includes the following alloys:

a. Plain carbon and low alloy steels heat treated to ultimate tensile strengths below 180 ksi
b. High strength martensitic steel (e.g., 4130, 4140, 4330V, 4340, 8630, 8740, 300M, 52100, M-50, etc.)
c. Secondary hardening steel (e.g., AerMet 100, AF1410, Ferrium M54/S53, etc.)
d. Maraging steel (e.g., Maraging 250/300/350, etc.)
e. Martensitic stainless steel (e.g., 403, 410, 416, 420, 431, 440C, Greek Ascoloy, etc.)

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f. Precipitation hardenable martensitic stainless steel (e.g., 15-5PH, 17-4PH, PH13-8 Mo, Custom 450/455/465, etc.)
g. Precipitation hardenable semi-austenitic steel (e.g., 17-7PH, PH15-7Mo, AM 350/355, etc.)
h. Precipitation hardenable austenitic steel (e.g., A-286)
i. Austenitic stainless steel (e.g., 201, 202, 301, 302, 303, 303 Se, 304, 304L, 316, 316L, 321, 347, etc.)

5.1.7.1 Material specifications for steel. Procurement of steel raw material or steels used in finished parts shall use SAE Aerospace Material Specifications (AMS). Unless otherwise authorized by the CEA, premium aircraft quality material shall be used for critical applications.

The material shall meet all technical requirements of the applicable AMS which defines the required manufacturing methods, quality assurance provisions and the properties shall be measured and met to assure that the product will be satisfactory for the intended usage. Acceptance tests for steel typically include composition, condition, macrostructure, micro-inclusion rating, average grain size, hardenability, decarburization, response to heat treatment, longitudinal and transverse tensile properties, nonmetallic inclusion content (cleanliness) and tolerances. Other material specifications (ASTM, Military Specifications, etc.) for steel may be used with approval from the CEA. Other specifications shall include similar types of acceptance tests as described above.

5.1.7.2 Heat treatment of raw steel material and steel parts. Raw material includes, but is not limited to, items such as sheet, plate, wire, rod, bar, forgings, and extrusions. Heat treatment of raw material shall be in accordance with SAE AMS2761. The heat treatment condition of raw materials shall be as specified in the procurement document or in the specific material specification for the alloy and product form. Heat treatment of steel parts (annealing, solutioning, normalizing, tempering, aging, carburizing, nitriding, etc.) shall meet the requirements of SAE AMS2759 and the applicable slash specification for the specific steel alloy and heat treatment process. Any deviation from these requirements or use of alternate heat treatment procedures shall be approved by the CEA.

5.1.7.3 Straightening during heat treatment. The design of the heat treatment processes for steels (heating equipment, racks, fixtures, quenchant, quench systems, etc.) shall be optimized to minimize warping and distortion during heat treatment while producing products meeting all metallurgical and mechanical property requirements. No impact bending, reverse bending, or localized bending is allowed. Allowances and limitations for straightening of steel parts during heat treatment and after heat treatment shall be in accordance with SAE AMS2759 and the applicable slash specifications. A straightening process with specified limits authorized by the CEA is required when straightening of any steel part after final tempering or precipitation hardening having a minimum ultimate tensile strength of 180 ksi or higher.

5.1.7.4 Machining of steel in the hardened condition. Steels heat treated to high tensile strength (i.e., in the hardened condition) can be difficult to machine without producing metallurgical damage (over-tempered/untempered martensite, over aging, smearing of metal surfaces, chatter marks, cracking, etc.). Steel parts that will be heat treated to 180 ksi ultimate tensile strength or higher shall be machined to near final size to the extent possible before hardening. High strength steel parts that will be machined after hardening shall have sufficient stock remaining after hardening to meet the removal requirements of SAE AMS2759 for

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contamination removal and decarburization control. Finished surfaces, such as threaded or unthreaded holes, that are protected during hardening by copper plating, copper plated plugs or other approved coatings do not require additional machining operations after hardening. New or alternative machining technologies for machining of hardened steels that produce machined surfaces free of metallurgical damage and defects may be used with verified and defined process controls, appropriate quality control provisions and CEA approval. When machining steel in the final hardened condition is necessary the following sections provide requirements for machining, including grinding and hole making.

5.1.7.4.1 Grinding. SAE AMS2453 shall be used for high strength martensitic steels, secondary hardening steels, and PH stainless steels heat treated to 180 ksi UTS or above as well as any of these alloys and heat treatments that are chrome plated. SAE AMS2449 shall be used for high strength steels with applied High Velocity Oxy-Fuel (HVOF) coatings. For the Air Force, Drawing 200310642: shall be used for high strength steels with applied HVOF sprayed Tungsten Carbide coatings. The grinding parameters of SAE AMS2453, SAE AMS2449 and Drawing 200310642 are intended to eliminate or minimize the risk of overheating, cracking, high residual tensile stresses, and other metallurgical changes that decrease structural integrity of steel parts, chrome plated steel parts or HVOF coated steel parts. Grinding process control sheets and part-specific inspection technique sheets as detailed in SAE AMS2453 or SAE AMS2449 shall be documented and used.

5.1.7.4.2 Grinding to MIL-STD-866 requirements. MIL-STD-866 may be used for high strength martensitic steels (e.g., 4140, 4330V, 4340, 300M, etc.). Grinding parameter sheets shall be used to record requirements for rigidity of the grinding setup, workpiece fixturing, spindle and quill requirements for internal grinding, grinding wheel type and size, the type of wheel dresser and the frequency of grinding wheel redressing, workpiece and grinding wheel speeds, traverse speeds, infeeds, etc. When grinding of high strength martensitic steels in accordance with MIL-STD-866 the following additional requirements apply:

a. Inspection of ground chrome plated parts shall be in accordance with SAE AMS2440.

b. Ground chrome plated parts showing possible rejection indications by fluorescent penetrant inspection or magnetic particle inspection may be cleaned and lightly sanded in accordance with the procedure and provisions of SAE AMS2440. Ground chrome plated parts rejected due to indications of grinding damage in accordance with SAE AMS2440 shall be stripped and the base metal surface shall be temper etch inspected in accordance with SAE AMS2649 or MIL-STD-867, magnetic particle inspected in accordance with ASTM E1444 and re-peened to specifications prior to re-plating.

5.1.7.4.3 Manufacture of holes. Manufacture of quality holes (including chamfering and spot facing) in hardened steels requires special precautions and precise control of the machining parameters to prevent metallurgical damage that can result in premature failure. Qualified hole manufacturing techniques (QHMTs) are required for all drilling, reaming, grinding, and honing processes performed on high strength steel parts that have been heat treated to 180 ksi minimum ultimate tensile strength or higher. The QHMTs shall contain the requirements described in sub-paragraphs a through g. Refer to 5.1.7.4.5 for qualification of hole manufacturing techniques.

a. Cutting fluids, coolants, or lubricants. The type of cutting fluid, coolant or lubricant is

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critical to hole manufacture and the exact composition, concentration range and maintenance requirements form a part of the hole manufacture technique qualification. Cutting fluids, coolants or lubricants shall be continuously applied during all metal removal operations.

b. Controlled feed. Controlled positive feed machine tools and equipment shall be used in all hole manufacture. The control parameters of the positive feed (feed rate, drill speed, etc.) shall be contained in the QHMT. Controlled tool feed is not required for manual reaming, honing, spot facing or chamfering.

c. Drilling. Carbide or high speed steel (HSS) drills with positive rake angles shall be used. Drill speed and feed rate shall be as specified in the qualified hole manufacturing technique (QHMT). Unless otherwise approved by the CEA, all holes drilled after hardening shall have a minimum of 3/64 inch (0.047 inch) metal removed on the diameter (0.024 inch on the radius) by reaming or reaming and honing to ensure removal of all surface contamination from hardening and metallurgical damage from drilling. Replace or re-sharpen drills at the interval specified in the QHMT or when there is any evidence of degraded performance (presence of chatter, poor surface roughness, change in sound or excessive heat).

d. Reaming. Carbide or high speed steel reamers are recommended. The specific reamers approved for use shall be documented in the QHMT. Reaming to final size may be accomplished by two step reaming (rough/final ream) or by reaming and honing or reaming and grinding so long as 3/64 inch on the diameter is removed and surface roughness and straightness requirements specified in the QHMT are met. Change-out or re-sharpen reamers at the interval specified in the QHMT or when there is any evidence of degraded performance (presence of chatter, poor surface roughness, change in sound or excessive heat).

e. Boring. Carbide, HSS, ceramics, and cubic boron nitride tools may be used for boring of hardened steels as specified in the QHMT. Machining to final size may be by boring or by boring and reaming, honing, or grinding is acceptable so long as a minimum of 3/64 inch (0.047 inch) on the diameter is removed. Change-out or re-sharpen boring tools at the interval specified in the QHMT or when there is any evidence of degraded performance (presence of chatter, poor surface roughness, change in sound or excessive heat).

f. Finishing holes by grinding. Grinding of holes to final dimensions can be used when other secondary hole making processes cannot meet hole dimensions or surface roughness requirements. High rigidity in the grinding machine and workpiece fixturing and frequent coarse dressing of the grinding wheel are critical factors in grinding of high strength steels without producing grinding burns. Slow wheel speeds, friable or semi-friable aluminum oxide grinding wheels with the largest grit size that can achieve the required surface roughness will also help produce low stress grinding conditions. The QMHT shall contain the grinding tools and parameters.

g. Other requirements or provisions. When honing or lapping of shot peened surfaces to meet surface roughness or hole tolerance requirements the maximum material removal from the peened surfaces shall meet the requirements of SAE AMS2430.

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5.1.7.4.4 Inspection and processing requirements during and after hole manufacturing. Visual and low power macroscopic examination of hole surfaces shall be performed after each hole manufacturing operation (e.g., drilling, boring, reaming or grinding), but before honing or lapping, for evidence of heat discoloration, surface tears, chatter marks, gouges, or any other evidence of abusive machining. Holes larger than 1/2 inch or having a suitable diameter to length ratio to permit "line of sight" examination shall be temper etch inspected in accordance with 5.1.7.5. Immersion or swab methods may be used. Surface roughness of the hole surfaces shall be measured in accordance with SAE AS291, ASME B46.1 or ASTM D4417, Method B. Surface roughness shall meet the requirements of the QHMT. Holes of 1/2 inch diameter or less or holes with a length to diameter ratio that precludes use of a temper etch inspections do not require etch inspection if manufactured in accordance with a QHMT. All high strength steel parts heat treated to 180 ksi or higher machined in the hardened condition shall be stress relieved in accordance with SAE AMS2759/11. Following stress relief, the parts shall be inspected by magnetic particle inspection in accordance with ASTM E1444 or fluorescent penetrant inspection (FPI) in accordance with ATSM E1417/E1417M. Parts rejected for evidence of excessive surface roughness, presence of heat discoloration, tears, gouges, cracks, rejectable temper etch indications, etc. shall be subject to engineering disposition prior to rework.

5.1.7.4.5 Hole manufacturing process qualification. Hole manufacture in steels heat treated to 180 ksi or higher by drilling, boring, reaming, honing, grinding, or combinations of these secondary operations, shall be performed using a qualified hole manufacturing technique (QHMT). The CEA shall determine if an existing QHMT is viable for use on other similar applications (i.e., similar alloy, heat treatment, part and hole configuration, and dimensions). To qualify a new hole manufacturing technique for CEA approval, the following requirements shall be performed:

a. Manufacture 20 holes of the required dimensions using the same hole making technique parameters (e.g., cutting fluids, drill speed, feed rate, cutting tools, drills, reamers, hones, grinding wheels, etc.) in specimens of the same steel alloy and heat treatment condition. Minimum tool change-out or re-sharpening frequency shall be identical to that used in development of the QHMT up to a maximum of 20 holes. Extended change-out or re-sharpening frequencies may be qualified by evaluation of successive holes made with the same tooling up to the desired change-out or re-sharpening frequency.

b. Visual and macroscopic examination and surface texture measurements of all specimens shall be performed. Hole surfaces shall be free from heat discoloration, tears, chatter marks, gouges, or any other evidence of abusive machining. Surface roughness shall meet the requirements of the QHMT for each separate machining method, as applicable.

c. Prepare the test specimen for metallographic examination. The test holes shall be sectioned longitudinally. Test specimens shall be temper etch inspected in accordance with 5.1.7.5. To evaluate for cracks, magnetic particle (in accordance with ASTM E1444) or fluorescent penetrant (in accordance with ASTM E1417/E1417M) inspections using a CEA approved procedure shall be performed. Test specimens may be further sectioned (if size requires) to sizes appropriate for metallographic mounting and for microscopic examination specifically at the tool entry point, mid-depth and at the tool exit or end point. Use metallographic techniques to ensure good

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edge retention during grinding and polishing. Remove sufficient material by grinding and polishing to remove any altered material caused by sectioning. After final polish etch with appropriate etchant, rinse and dry.

d. Microscopically examine the specimens at magnifications up to 500X along the hole surfaces. Examination of all specimens shall show no untempered martensite, cracks, excessive plastically deformed surface layers, presence of microstructural changes due to overheating (e.g., presence of over-tempered martensite or untempered martensite, phase changes, etc.), or other objectionable defects.

5.1.7.5 Temper etch inspection. Temper etch inspection is used to detect the presence of over-tempered or un-tempered martensite on the surfaces of parts resulting from overheating during grinding (grinding burns) or other machining operations. Temper etch inspection may also be used to detect incomplete carburization, localized decarburization, or unauthorized carburization. Temper etch inspection is only used on high strength martensitic steels (e.g., 4140, 4340, 300M, etc.). It is not applicable to stainless steels or any steel that is not effectively etched by nitric acid solutions. All high strength martensitic steels hardened (quenched and tempered) to an ultimate tensile strength of 180 ksi or higher that are ground shall be temper etch inspected in accordance with SAE AMS2649 or MIL-STD-867 after final grinding and before shot peening, plating, coating, or any other process that alters or masks the ground surface. Non-embrittling methods such as SAE AMS2649, type 1 or MIL-STD-867, table V are highly recommended for temper etching. Total immersion of parts in the etching solutions results in the most uniform and consistent etching especially for large surface areas and is mandatory except for very small areas or special circumstances where immersion etching cannot be performed due to fluid entrapment or when in-situ (e.g., on aircraft) inspections are necessary. In these cases, swab etching using either the non-embrittling JAR-3N additive (or equivalent) methods or the Ammonium Persulfate method may be used.

5.1.7.6 Hydrogen embrittlement. Certain high strength steel parts heat-treated to strength ranges identified in SAE AMS2759/9 are susceptible to hydrogen embrittlement (HE) from cleaning, electroplating, or other processes where hydrogen is evolved. Hydrogen embrittlement relief baking (de-embrittlement) of high strength steel parts shall be in accordance with SAE AMS2759/9. The steels, steel parts, heat treatment conditions and the potentially embrittling plating processes and chemical treatments requiring hydrogen relief baking are specified in SAE AMS2759/9.

5.1.7.7 Barkhausen noise inspection for grinding or machining burns. Barkhausen noise inspection (BNI), in accordance with SAE ARP4462, for the detection of over-tempered or un-tempered martensite in high strength martensitic steels (e.g., ferromagnetic steels) caused by grinding or other machining operations has the following limitations and shall not be used unless specifically authorized by the CEA:

a. BNI suffers a loss of sensitivity on surfaces that have residual compressive stress (i.e., surfaces subjected to blast cleaning, shot peening, etc.). High strength martensitic steels are frequently mechanically cleaned (glass bead or aluminum oxide blasted) during manufacture and repair and are almost always shot peened especially on surfaces to be plated or coated.

b. BNI suffers a loss of sensitivity in the presence of nickel plating. BNI through ground

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chrome plate is limited by the thickness of the plating. Disposition of chrome plated surfaces rejected by BNI requires stripping of the chrome plate and temper etch inspection in accordance with SAE AMS2649 or MIL-STD-867.

c. BNI inspection of large surface areas requires a compatible BN analyzer and sensor, appropriate sample and sensor holding, and sample and sensor movement apparatus. In contrast, temper etch inspection by immersion methods offers rapid inspection of large surface areas.

d. Measurements near surface edges are difficult. In some cases, the effective measurement area cannot be moved entirely to the edge of a surface while also keeping both magnetizing poles in contact with the surface. The edges of ground surfaces and edges of ground plated surfaces are often the areas where most grinding damage occurs.

5.1.8 Titanium and titanium alloys.

5.1.8.1 Material specifications for titanium and titanium alloys.

a. Titanium bar, forgings, wire, and rings specifications. All titanium rod, bar, and forgings shall be in accordance with SAE AMS4921, SAE AMS4970, SAE AMS6900, SAE AMS6901, SAE AMS6905, SAE AMS6906, SAE AMS6907, SAE AMS6910, SAE AMS6915, SAE AMS6920, SAE AMS6921, SAE AMS6925, SAE AMS6926, SAE AMS6930, SAE AMS6931, SAE AMS6932, SAE AMS6935, SAE AMS6936, or SAE AMS6940 as applicable.

b. Sheet, strip, and plate. Titanium sheet, strip, and plate, as applicable, shall be in accordance with ASTM B265, SAE AMS4900, SAE AMS4901, SAE AMS4902, SAE AMS4903, SAE AMS4904, SAE AMS4907, SAE AMS4909, SAE AMS4910, SAE AMS4911, SAE AMS4915, SAE AMS4916, SAE AMS4917, SAE AMS4918, SAE AMS4919, SAE AMS4939, SAE AMS4940, SAE AMS4988, SAE AMS4989, or SAE AMS4990.

c. Extrusions. All titanium extruded bars, rods, or special shaped sections shall be in accordance with MIL-T-81556. Other material specifications and product forms require approval by the CEA.

e. Welded or seamless titanium tubing. Welded or seamless hydraulic titanium tubing shall be in accordance with SAE AMS4941, SAE AMS4942, SAE AMS4943, SAE AMS4944, or SAE AMS4945 as applicable.

5.1.8.2 Titanium raw material and parts. Titanium raw material and parts shall meet the melting practice, product form, composition, heat treatment, properties, and quality requirements of the SAE AMS for the specific alloy and product form required by the design. Heat treatment temperatures, dwell times, cooling rates, etc. shall be as specified by the material specification. When not specified in the material specification, SAE AMS2801 requirements (e.g., all furnace equipment, furnace atmospheres, auxiliary equipment and quenching facilities and media) shall be used.

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5.1.8.3 Titanium wear. Titanium and its alloys exhibit very poor wear resistance and fretting can contribute to crack initiation, especially fatigue initiation, when the protective oxide layer is disrupted, resulting in corrosion. Refer to MIL-STD-1568 for recommendations for use of anti-fretting and anti-galling compounds on titanium parts or surfaces subjected to fretting.

5.1.8.4 Titanium flammability. Titanium alloys shall not be used with liquid or gaseous oxygen at any pressure or with air at oxygen partial pressures above 35 kPa (5 psia) due to the potential for titanium fire.

5.1.9 Nickel based and cobalt based alloys.

5.1.9.1 Heat treatment. Heat treatment of nickel and cobalt alloy parts shall be in accordance with SAE AMS2773, SAE AMS2774 or as specified by the CEA. When required by the CEA, critical parts shall require verification of tensile properties by use of process control test specimens in addition to other CEA specified tests (e.g., hardness, conductivity, etc.).

5.1.9.2 Cleaning. Cleaning of nickel and cobalt parts before, during and following manufacture shall be in accordance with MIL-DTL-5002. Uncoated nickel alloys are susceptible to sulfur embrittlement at high temperatures and require any foreign material that could contain sulfur, such as oils, grease, and cutting lubricants, to be removed by specified methods prior to heat treatment, elevated temperature exposure, welding, or high temperature service.

5.1.9.3 Machining. Nickel and cobalt superalloys are typically difficult to machine because of low thermal conductivity, high hardness, and severe work hardening resulting in high cutting forces, high cutting temperatures, severe tool wear, and poor machined surface quality. The high temperature, high stresses and high strain rates can result in drastic changes (damage, severe plastic deformation, etc.) in the microstructure at the workpiece surface. This surface damaged layer is called a white layer (a generic term for a hard layer that exists on the surface or subsurface of a metallic material that has no characteristic morphology and appears white under a light microscope after etching). The presence of this machining damage (white layer) can result in loss of mechanical properties, especially fatigue, wear resistance and SCC resistance. Machining processes, and post machining inspections of nickel and cobalt alloys to eliminate the detrimental effects of surface and near surface defects including white layer, shall be approved by the CEA.

5.1.10 Magnesium alloys. Magnesium alloys shall not be used as primary aircraft structure or in other areas that are subject to wear, abuse, foreign object damage, abrasion, erosion, or where moisture entrapment is possible unless specifically authorized by the CEA. Magnesium alloys shall not be ground, machined, or sanded inside aerospace weapons systems during ground processing due to possible ignition of magnesium chips and powders.

5.1.10.1 Heat treatment of magnesium alloys. Heat treatment of magnesium alloy castings shall be in accordance with SAE AMS2768. When SAE AMS2768 is specified and specific heat treat processing instructions are included in the material specification, the equipment and control requirements of SAE AMS2768 shall apply. For alloys not specifically covered, the applicable provisions of this specification shall be invoked but temperatures, times, and quench procedures shall be as specified by the cognizant engineering authority (CEA). Heat treatment of wrought magnesium raw material or parts shall be in accordance with ASTM B661.

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5.1.11 Beryllium and beryllium alloys. Beryllium alloys and oxides of beryllium shall not be machined inside the crew compartments of aerospace weapons systems at any stage of manufacturing, assembly, testing, modification, or operation.

5.1.11.1 Beryllium alloys containing more than 4 percent by weight beryllium. These alloys are exceptionally lightweight but have limited applications (outside of electronics, sensors, etc.) due to the extreme toxicity of beryllium salts and oxides and the high susceptibility to physical and mechanical damage. The following restrictions apply:

a. Alloys containing more than 4 percent beryllium by weight shall not be used for primary structural applications.

b. Alloys containing more than 4 percent beryllium by weight shall not be used for any application within aerospace crew compartments unless suitably protected to prevent erosion or formation of salts or oxides and authorized by the CEA.

c. Design of beryllium parts for aerospace weapons systems parts that provide mission critical functions shall include an assessment of the low-impact resistance and notch sensitivity (particularly at low temperatures), the significant anisotropy and sensitivity to surface roughness requirements for the alloy under consideration.

d. Beryllium alloy parts (>4 percent Be) used in aerospace weapons systems shall be processed to ensure complete removal of the damaged surface layer (twins and micro-cracks) produced by machining and grinding. Chemical milling and etching are effective processes for removal of the damaged layer.

5.1.12 Other metals and alloys. Other metals and alloys may be used in aerospace weapons systems designs with the approval of the CEA.

5.1.13 Tubing. Requirements for standard hydraulic tube walls, tube bending, and damage limits shall be in accordance with SAE AS4467, SAE MA2008 or SAE AIR1388 as applicable. Requirements and standards for impulse testing of hydraulic tubing shall meet SAE AS4265 and SAE AS603. Flexure testing procedures and best practices shall meet SAE AIR1529 and SAE MA2003. Burst testing on hydraulic tubing shall meet standards SAE AS85421 or SAE AS85720.

5.2 Nonmetallic materials. Nonmetallic materials shall be selected and used in compliance with the requirements contained in the following subparagraphs.

5.2.1 Composites. The selection of materials and processes for composites shall consider all aspects of the intended application, such as service environment, systems requirements, structural and functional requirements, serviceability and repairability, etc. Composite materials shall be evaluated in accordance with ASTM, SAE, CMH-17 or other relevant government or industry specifications as approved by the CEA. Training requirements are detailed in SAE AIR 4938 for composite and bonding repair artisans and SAE AIR6825 for composite/bonding inspectors.

5.2.1.1 Organic matrix composites. Materials and processes for organic matrix composites shall be in accordance with SAE CMH-17-1, SAE CMH-17-3, NAVAIR 01-1A-21 and

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NAVAIR 01-1A-22, Air Force TO 1-1-690, Army TM 1-1500-204-23-11 or equivalent as approved by the CEA. Guidance in the processing and production of high performance organic composite materials and structures can be found in the DoD/NASA Structural Composites Fabrication Guide, Volume I and Volume II. Guidance in the effective utilization of high performance organic composite materials and design concepts in aerospace structures can be found in the DoD/NASA Advanced Composites Design Guide, Volume I - Volume IV, and in SAE CMH-17-1, SAE CMH-17-3, NAVAIR 01-1A-21, and NAVAIR 01-1A-22.

5.2.1.2 Metal matrix composites. Metal matrix composites are comprised of a metal matrix (i.e., aluminum alloys, magnesium alloys, titanium alloys, etc.) where the metal provides more of the load carrying capability than the organic binder of an organic matrix composite. SAE CMH-17-4 shall be used as guidance for the materials and processes of metal matrix composites.

5.2.1.3 Ceramic matrix composites. Ceramic matrix composites are comprised of a ceramic matrix (i.e., carbon, silicon carbide, alumina, etc.) where the ceramic matrix allows for higher temperature applications versus the organic binder of an organic matrix composite. SAE CMH-17-5 shall be used as guidance for the materials and processes of ceramic matrix composites.

5.2.1.4 Composite material property design allowables. In addition to the requirements in 4.6, the effects of variations in processing upon the design allowables for conditions such as fiber (ply) mis-orientations, temperature and pressure variations, non-uniform chopped fiber distribution, thermoplastic morphology, etc. shall be evaluated. Guidance on composite material properties can be obtained from SAE CMH-17 handbooks.

5.2.1.5 Raw materials composites. Raw materials used for the fabrication of composites shall be labeled for storage with the date received, batch or lot number, name, specification or procurement document number, unique storage requirements, requirements for documenting raw material out time and out life, shelf-life expiration, etc. In addition, re-qualification procedures for expired materials shall be specified, if applicable. See the FAA Advisory Circular AC 43-214A “Repairs and Alterations to Composite and Bonded Aircraft Structure” for additional guidance.

5.2.1.6 Sandwich assemblies. Honeycomb or foam core sandwich assemblies shall be designed and fabricated to preclude the accumulation and entrapment of water or other contaminants within the core structure. Post assembly edge sealing shall be used in addition to design techniques to preclude liquid entry. Perforated metallic honeycomb core shall not be used. Aluminum honeycomb core shall be in accordance with SAE AMS-C-7438 and shall be of the corrosion resistant type. Phosphoric acid anodized (PAA) aluminum core coated with corrosion inhibited primer is preferred for sandwich assemblies at risk for damage or corrosion. Sandwich construction using non-metallic honeycomb core and composite facings shall be in accordance with MIL-S-9041 except 3.3 and table I therein shall not apply. MIL-S-25392 shall apply for construction using foamed-in-place core and uniform composite facings and SAE AMS-C-8073 shall apply for non-metallic honeycomb core and fiberglass facings. Design of structural sandwich assemblies shall be guided by SAE CMH-17-6. The design shall be validated by tests typical of the end-use environment. They shall include vibration and acoustic testing. Other core materials or designs, for which the above specifications do not apply, require approval of the CEA.

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5.2.1.7 Joining and fastening. Potential mechanical damage to the composite holes from repeated insertion and removal of fasteners shall be evaluated. SAE CMH-17 handbooks provide information about joining, hole quality, and fastener installation that shall be controlled by detailed process specifications.

5.2.1.7.1 Thermal expansion. The effect of residual stresses, introduced by cool-down from some stress-free elevated temperature existing during cure, shall be evaluated in design. Particular attention shall be paid to thermal expansion mismatch in evaluation of any manufacturing process that employs multiple elevated temperature cure cycles. Any elevated temperature cure (including secondary bonding and post-curing operations) shall not result in degradation of bondline integrity, either for load transfer or sealing purposes. Aluminum fittings, etc., shall not be bonded in conjunction with graphite composites, unless demonstrated to be thermally compatible by a CEA approved thermal stress analysis procedure.

5.2.1.7.2 Galvanic corrosion. Since corrosion of most metals in contact with graphite composites will be accelerated, these couplings shall be avoided. Particular attention shall be paid to sandwich structure using metallic core with carbon skins and potential mitigation approaches such as thick adhesive bondlines, fiberglass isolations plies, etc. When this coupling cannot be avoided, it shall be treated as dissimilar materials and protected in accordance with the requirements of MIL-STD-889, MIL-STD-7179, and MIL-STD-1568.

5.2.1.8 Repairability. The selection of composite repair materials and processes, identification of appropriate standardized composite repair methods, and engineering design of composite repairs shall be in accordance with NAVAIR 01-1A-21, NAVAIR 01-1A-22, AF T.O. 1-1-690, TM 1-1500-204-23-11, AFWAL-TR-83-3092, SAE CMH-17-3, or equivalent, as approved by the CEA. For composite repair materials and processes not covered by these documents, the OEM shall identify, develop, and provide detailed repair methods and procedures including ply level details (e.g., number of plies, ply orientations, ply material, ply drops/adds, core direction, etc.) for CEA approval. For Air Force weapons systems, T.O. 1-1-690 should be used to cover current shop best practices for executing the repairs.

5.2.2 Elastomeric materials.

5.2.2.1 General requirements. All elastomeric components shall be reversion resistant, hydrolytically stable, and possess the necessary resistance to aging, operational environmental conditions, and fluid exposure required for the intended systems use as determined by the CEA.

5.2.2.1.1 Cured elastomers. Cured elastomers that are age sensitive shall be in accordance with SAE AS1933 and SAE ARP5316. All cured elastomeric materials shall have the cure date either on the item itself or on the packaging. A policy of first in, first out shall be maintained. Cured elastomeric materials shall be protected from UV radiation, fuel, oil, water, dust, and ozone (which is generated by electric arcs, fluorescent lamps, and similar electrical equipment). The storage temperature of cured elastomers shall not exceed 55 ºC (125 ºF). For optimum storage life, the storage temperature shall not exceed 38 ºC (100 ºF).

5.2.2.1.2 Non-cured elastomers. Materials that are procured in a non-cured state, such as sealants and potting compounds, shall be held in controlled temperature storage that does not exceed 26 ºC (80 ºF). Some specific materials may require storage at reduced temperatures and these materials shall be stored as recommended by the manufacturer. Materials requiring reduced

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temperature storage shall be avoided because of the added burden on reduced temperature storage and the likelihood of reduced temperature storage not being maintained at all times. Adequate storage times shall be set up and maintained. Most polysulfide, polythioether, and polyurethane sealants can be stored for at least nine months at less than 26 ºC (80 ºF) and the materials remain suitable for use. A policy of first in, first out shall be maintained. Uncured elastomers with an expired shelf life shall be discarded unless specific tests approved by the CEA demonstrate that the shelf life can be extended safely.

5.2.2.1.3 Silicone elastomers. Unless otherwise specified by the CEA, non-corrosive one-component silicone sealants that liberate alcohol during cure, in accordance with MIL-A-46146, shall be used for aerospace weapons systems requiring silicone sealant. Sealants for electrical applications shall be in accordance with SAE AS12500. Due to the properties of silicone sealants, caution shall be exercised to prevent silicone cross-contamination with other operations/processes. One-part silicone sealants that liberate acetic acid shall not be used unless approved by the CEA. When sealants that liberate acetic acid, such as MIL-A-46106, are approved for use, the following requirements shall be enforced:

a. Good ventilation during cure.
b. Thickness limit of ¼ inch, maximum.
c. Maximum bond line limit of 1 inch, when used between nonporous surfaces.
d. Sufficient moisture to complete cure.
e. Full cure before enclosure (7 days, minimum).

5.2.2.2 O-rings. Dimensions and tolerances for all O-rings shall be in accordance with SAE AS568. O-rings conforming to table II shall be used.

5.2.2.3 Other elastomeric parts. Other molded parts, sheets, strips, and extruded shapes shall be fabricated from the elastomeric material specifications listed in table III.

5.2.2.4 Potting compounds. The potting compounds shall be in accordance with the specifications listed in table IV.

5.2.2.5 Integral fuel tank sealing. Integral fuel tanks shall be sealed using sealants conforming to SAE AMS-S-8802, SAE AMS3276, SAE AMS3277, SAE AMS3278, SAE AMS3279, SAE AMS3281, and SAE AMS-S-83318. When non-curing groove injection type sealing is used, the material shall be beaded in accordance with MIL-S-85334 (or equivalent as approved by the CEA), SAE AMS3376, or SAE AMS3283. Adhesion promoter in accordance with SAE AMS3100, or as recommended by the sealant manufacturer, shall be used prior to sealing over SAE AMS-C-27725 polyurethane coating. The use of an adhesion promoter is advantageous when applying curing type sealant over polyurethane coating conforming to SAE AMS-C-27725 in accordance with MIL-STD-7179. While the sealant will adequately adhere to a new polyurethane coating, as the coating becomes older, obtaining proper adhesion becomes more difficult. This is true not only with fuel aged coatings, but also with coatings that have only been subjected to air.

5.2.2.6 Environmental sealing. For general sealing of aerospace weapons systems (other than fuel tanks and bladders) the sealants and procedures used shall meet the requirements of MIL-STD-1568 and MIL-STD-7179.

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5.2.3 Fluorocarbon seals. Fluorocarbon seals in accordance with SAE AMS7276, SAE AMS7259, SAE AMS7287, SAE AMS7410, or SAE AMS7379 shall be used.

5.2.4 Foamed plastics. Foamed plastics shall be hydrolytically stable when tested according to the CEA approved methodology. Foamed plastics can absorb moisture when exposed to humidity, water, or other liquids. Polyester polyol based polyurethane foams lack such stability and shall not be used in a moisture containing environment. The design of foam core sandwich or other constructions shall provide complete sealing against exposure to humidity and to fluids. The design shall be tested according to CEA approved methodologies to ensure sealing and resistance to vibration and acoustic noise meet the contract requirements.

5.2.5 Flammability. All non-metallic materials shall meet federal, state, and local flammability requirements as well as the performance requirements for the intended use.

5.2.5.1 Flammability requirements for interior compartment materials. The provisions of FAR 25.853 or other Federal Aviation Regulations shall apply in the design and selection of aircraft cabin interiors. Amendment 25-61 to FAR 25.853 requires improved flammability performance for aircraft cabin interior materials with large outer surface areas. Background information for the design and selection of materials for aircraft cabin interiors to minimize fires and emission of smoke and toxic fumes is found in JSSG-2010. Any material that is used in aircraft crew interior design that has not been certified by the FAA testing requirements shall be self-extinguishing and meet the CEA approved test parameters and criteria. For upholstery fabric, only self-extinguishing materials shall be used that meet the flame resistance and smoke generation requirements of the respective material types, as specified in SAE AMS3856. Carpeting materials used in manned aircraft, tested in accordance with ASTM E162, shall have an average Radiant Panel Index (Is), rounded to the nearest multiple of 5, of 65 or less. Any cabin furnishings, upholstery fabrics, and carpeting materials used that contain wool shall be treated with fire retardants. Polyvinyl chloride (vinyl or vinyon), modified aramid (durette), and phosphorous-based fire-retardant treated cotton shall be prohibited due to the toxic hazard level of the thermal decomposition products. Polyvinyl chloride-coated fabrics shall also be prohibited due to the toxic hazard level of thermal decomposition products. Materials in fabric category for usage such as curtains, coated fabrics, insulation covers, outermost seat coverings, headliners, nonwoven, and thermal barriers shall be self-extinguishing, and if not certified to FAA testing requirements shall meet the requirements of 25 CFR, Section 25.853, and appendix F, Part 1(b)(3). CEA approval is required for any deviations or changes to the test procedures or acceptance requirements.

5.2.5.1.1 Aircraft seats. Materials used for aircraft seats shall pass the following test(s): the outermost covering used to cover ejection seats shall be tested in accordance with 25 CFR, Section 25.853 and appendix F, Part 1(b)(3) and shall comply with the CEA approved test parameters and criteria. Non-ejection crew and passenger seating shall pass the flame impingement test prescribed in FAR Part 25.853, Amendment 25-59. Fire blocking layers between the outermost seat upholstery and cushions (seat back and bottom) may be required by the CEA to meet FAR Part 25.853 requirements. The outermost seat covering for all seats shall pass their respective flame retardancy test requirements before and after 10 dry cleanings in accordance with 25 CFR, Section 25.853 and appendix F, Part 1(b)(3). CEA approval is required for any deviations or changes to the test procedures or acceptance requirements. Test samples shall be determined individually as well as the average value.

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5.2.5.2 Fire insulating paint. Within power-plant compartments of aircraft and other compartments normally operating at temperatures below 300 °F (149 °C), where fires are likely to occur as a result of flammable fluid leakage, and in areas adjacent to bleed air ducts and valves that contain air at temperatures above 300 °F (149 °C), all fluid containers (air bottles, oxygen containers, hydraulic reservoirs, accumulators, and cylinders) that could escalate the intensity of a fire by explosion due to excessive heat, shall be protected by a finish system. The finish system shall consist of a primer qualified and applied in accordance with MIL-PRF-23377, MIL-PRF-85582, or other CEA approved primer followed by a thermally insulating fire barrier topcoat and sealants (such as SAE AMS3374 or SAE AMS3381) as approved by the CEA.

5.2.6 Lubricants and working fluids.

5.2.6.1 Lubricants. Molybdenum disulfide based anti-seize compounds such as MIL-PRF-83483 shall not be used in applications where temperatures exceed 410 ºC (800 ºF). When polytetrafluoroethane (PTFE) and similar materials are used as a self-lubricating surface, such as on wing pivot fittings, bearing races, and other applications, the design shall be based on demonstrating wear life in the presence of the fluids likely to be in contact with the wear surfaces. Lubricants or anti-seize compounds containing graphite shall not be used, except for use on aircraft engine spark plugs and threaded fasteners and fittings where temperatures are expected to be above 410 ºC (800 ºF). Solid film lubricants shall not be employed on the internal surfaces of hydraulic or fuel systems. Lubricants containing sulfur shall not be used in contact with gold or silver due to the formation of metal sulfides which can negatively impact applications such as connectors. MIL-HDBK-838 and MIL-HDBK-275 shall be used for guidance on the selection and use of lubricants.

5.2.6.2 Working fluids. Working fluids such as hydraulic fluids, coolants, and heat transfer fluids shall be approved by the CEA. MIL-HDBK-838 and MIL-HDBK-275 shall be used for guidance on the selection and use of these working fluids.

5.2.7 Transparencies. Requirements for transparencies shall be identified and approved by the CEA. The following specifications may be used by the CEA as applicable: JSSG-2006, JSSG-2010, MIL-HDBK-516, MIL-PRF-25690, MIL-PRF-8184, SAE AMS-P-83310, ASTM D1003, ASTM F733, and ASTM F1165. Aerospace transparencies are used in a wide variety of applications (canopies, windows, lens, goggles, sensors, etc.) and thus the mechanical and physical properties vary for each specific application. Material type and thickness are based on mechanical requirements (impact capability, escape system requirements, cabin pressure, abrasion resistance, chemical resistance, etc.). Optical requirements for transparencies and transparency coatings (light transmission [visible and NVIS], haze, distortion, angular deviation, binocular vergence, etc.) also vary widely depending on the application and the electromagnetic spectrum wavelengths involved. In some cases, a single transparency may have several different optical requirements, or zones, based on the application or the task the pilot is performing when looking through those different areas. Detailed design requirements for the broad applications and required material properties are beyond the scope of this standard.

5.2.8 Electrical insulation. Wiring insulation materials and installation procedures as described in SAE AS50881 shall be used to ensure long term insulation performance. Polyimide insulation (MIL-STD-81381) shall not be used in U.S. Department of Defense contracts for aerospace applications. Vinyl and polyvinylchloride material shall not be used as electrical insulation or sleeving. Vinyl and polyvinylchloride material supports fungus growth. Satisfactory

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materials for wire insulation include ethylene-tetrafluoroethylene (ETFE), polytetrafluorethylene (PTFE), fluorinated ethylene propylene (FEP), Kel-F, polyurethane, polycarbonate, polyethylene, polyalkene, polyethylene terephthalate, polyolefin, polysulfone, and silicone sleeving. Polyimide-insulated copper does not meet minimum performance requirements due to arc tracking, fluid degeneration, and embrittlement. The fungi produced release vapors that are known to be aggressive to metals, plastics, and elastomers. Outgassing of this growth will occur under nominal storage conditions but will be accelerated in both high temperature and low-pressure environments. Exercise caution in the use of PTFE covered silver plated copper wire because of possible corrosion at pin holes. Good adhesion is difficult to obtain when potting or encapsulating PTFE insulated wire. Wire coated with PTFE and FEP may "cold flow" when installed in sub-optimal conditions. Sub-optimal conditions largely include general stress, against sharp edges, and in sharp bends that result in shorting failures.

5.2.9 Tape. Tapes used on interiors of aerospace weapons systems shall be in accordance with CEA requirements and be non-corrosive and non-moisture absorbing in addition to having resistance to fungal growth and no out-gassing. Tapes shall not be used in exterior areas without the approval of the CEA.

5.2.10 Non-hygroscopic materials. Non-wicking, non-hygroscopic gaskets shall be used to prevent moisture intrusion. Felt, leather, cork, or glycol impregnated gaskets shall be avoided as well as cotton core material in electrical cables. Asbestos shall not be used. The outer edges of hygroscopic laminated assemblies shall be sealed to prevent moisture intrusion.

5.2.11 Water displacing compounds. Water displacing compounds, in accordance with MIL-STD-7179, shall be used to protect metal surfaces against moisture, fingerprints, and corrosion. On plated surfaces or electrical devices including leads, contacts, and terminal posts, the soft film types of such compounds have been found to be effective protection against corrosion at pores or pinholes in the protective plating, a defect frequently found with standard commercial items. The water displacing compounds shall be in accordance with applicable military specifications or non-government standards. Other corrosion preventive compounds shall be approved by the CEA.

5.2.12 Moisture and fungus resistance. Parts and equipment shall be designed so that the materials are not nutrients for fungi, except when used in permanent hermetically sealed assemblies and other accepted and qualified parts such as treated transformers. Other necessary fungi nutrient material applications shall require treatment by a method that will render the resulting exposed surface fungi resistant. The criteria for the determination of fungi and moisture resistance shall be in accordance with MIL-STD-810.

5.3 Manufacturing processes.

5.3.1 Near-net shape forging. To the maximum extent possible, forgings shall be designed to have an internal grain flow pattern that is predominately parallel to the axis of principal stresses. When this is not possible, the forged part shall be designed such that the weakest grain flow direction (ST direction) is not parallel to the axis of principal stresses. Forgings shall also be designed to be free from reentrant and sharply folded grain flow lines. When the CEA determines first article qualification of the forging is required, preproduction qualification of forgings and the controls to be used for follow-on production forgings shall be in accordance with SAE AMS2375 and the following requirements:

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a. Preproduction qualification (first article approval) of critical and semi-critical forgings used in aerospace weapons systems shall be approved by the CEA. Deviations from the requirements of SAE AMS2375 require CEA approval.

b. The mechanical properties (grain flow, hardness, microstructure, tensile properties, etc.) of the forging control coupons taken from within first article forging at specified locations shall be correlated to the results from prolongations that will be used to verify production forgings to show they are predictive of the properties in the body of preproduction forgings and can be used for quality control of production forgings. Final coupon to component correlation used for quality control and assurance shall be approved by the CEA.

c. Prolongations shall be obtained for each critical or semi-critical production forging and tested for required minimum mechanical properties to ensure the mechanical properties of each forging is achieved. The orientation of the prolongation shall represent the direction for which the specified mechanical properties are required.

d. A process control factor plan for each forging shall be approved by the CEA. Production forgings shall be produced using the same operations, practices, and process control factors as those used on the approved preproduction forgings. A change to any of the forging process control factors for producing forgings specified in SAE AMS2375 shall require new preproduction (first article) verification unless otherwise authorized by the CEA.

e. Surface and volumetric nondestructive inspection (NDI) shall be performed on all critical and semi-critical forgings using the methods, criteria and procedures specified for the forging.

5.3.2 Castings. Castings used in critical and semi-critical aerospace weapons systems shall meet the requirements of SAE AMS2175 and the following:

a. Preproduction qualification approval (first article approval) of critical and semi-critical castings used in aerospace weapons systems shall be approved by the CEA. Deviations from the requirements of SAE AMS2175 require CEA approval. Preproduction castings used for first article approval shall have the same process control factors as the production castings.

b. The mechanical properties (hardness, microstructure, tensile properties, etc.) of the casting control coupons taken from within the first article casting at specified locations shall be compared to the results from prolongations, trim material or integrally cast specimens that will be used to verify production castings to show they are predictive of the properties in the body of preproduction casting and can be used for quality control of production castings. Final coupon to component correlation used for quality control and assurance shall be approved by the CEA.

c. For critical and semi-critical castings (class 1 and class 2 castings defined in SAE AMS2175) mechanical property testing of integrally cast or excised tensile bars at critical locations shall be conducted to ensure foundry control of cast lots.

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d. A process control factor plan for each casting shall be approved by the CEA. Production castings shall be produced using the same operations, practices, and process control factors used on the approved preproduction castings. A change to any of the casting process control factors for producing castings shall require new preproduction (first article) verification unless otherwise authorized by the CEA.

e. Periodic destructive testing of critical and semi-critical production castings for verification of tensile properties and to ensure the casting process is under control and reproducible may be specified by the CEA.

f. Surface and volumetric nondestructive inspection shall be performed as specified in SAE AMS2175.

5.3.3 Fusion welding. The methods, procedures and quality assurance requirements for manual, automatic, and semiautomatic welding for DoD aerospace weapons systems shall be in accordance with AWS D17.1/D17.1M and the following modifications/additions:

a. Critical welds shall comply with AWS D17.1/D17.1M Class A weld requirements.

b. Semi-critical welds shall comply with AWS D17.1/D17.1M Class B requirements.

c. Non-critical welds shall comply with AWS D17.1/D17.1M Class C requirements.

d. All Class A and Class B welds (including manual welds), as defined by AWS D17.1/D17.1M, shall be qualified in accordance with the procedures and requirements of AWS D17.1/D17.1M.

e. Fusion welding of titanium and titanium alloys should use AWS G2.4/G2.4M for guidance. Surface coloration of titanium welds after welding shall be light/dark straw or better in accordance with AWS D17.1/D17.1M indicating acceptable prevention of oxygen absorption.

f. Titanium and its alloys shall be welded with alloy-matching or metallurgically compatible fillers or autogenously.

g. Extra low interstitial (ELI) filler wires shall be used for titanium cryogenic applications and are preferred for general applications.

h. Commercially pure (CP) titanium filler shall not be used on Ti-6Al-4V or other alloyed base material to avoid hydride formation and embrittlement.

i. Nitrogen, hydrogen, carbon dioxide, and mixtures containing these gases shall not be used in welding titanium and titanium alloys. The inert gases of argon or helium shall have a dew point of -76 ºF (-60 °C) or lower.

j. Welded alpha and alpha plus beta titanium alloys shall be stress relieved in a vacuum or inert gas environment (argon or helium) or stress relieved in air followed by verification of oxide removal per SAE AMS-H-81200 or SAE AMS2801 as specified in the design.

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k. Laser welding of critical and semi-critical aerospace parts shall be in accordance with AWS D17.1/D17.1M and AWS C7.4/C7.4M.

l. Electron beam welding of critical and semi-critical aerospace weapons systems parts shall be in accordance with SAE AMS2680.

m. Welding of aerospace weapons systems parts or components to “non-aerospace” parts or components shall be in accordance with specifications and procedures mutually agreed to by the respective CEAs.

n. The following is not permitted:
   1. Welding from both sides if full penetration of the first pass is not verified by either inspection of the back side or grinding prior to welding the opposite side.
   2. Partial weld penetration in structural welds.
   3. Straightening operation after welding.

o. The Welding Procedure Specification (WPS) shall include the following additional content:
   1. Prequalified rework welds per AWS D17.1/D17.1M.
   2. Testing and documentation of allowable parameter variations for automatic and semi-automatic welds.
   3. Manual welding parameters.
   4. An associated Procedure Qualification Record (PQR) with tension testing and macro-examination results as part of weld qualification requirements.

5.3.4 Resistance welding. Resistance spot and seam welding of critical and semi-critical aerospace weapons systems hardware shall be in accordance with AWS D17.2/D17.2M. Resistance welding is the only approved process for fusion welding of clad high strength aluminum alloys.

5.3.5 Friction-stir welding of aluminum alloys. Friction-stir welding of aluminum alloys of critical and semi-critical aerospace weapons systems parts shall be in accordance with AWS D17.3/D17.3M.

5.3.6 Rotary friction welding. Rotary friction welding of aerospace weapons systems critical and semi-critical parts shall be in accordance with AWS C6.2/C6.2M and the following additional requirements:

a. Surface inspection by FPI and volumetric inspection by x-ray radiography shall be performed.

b. All welds shall be proof tested unless waived by the CEA.

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c. Inertia welds used in fluid systems shall be helium leak tested per system requirements.

5.3.7 Brazing. Brazing of critical or semi-critical aerospace weapons systems parts shall be in accordance with AWS C3.3 and the following additional requirements as applicable:

a. Critical and semi-critical brazed joints shall be designed for shear loading and shall not be used for axial loading of structural parts. The shear strength of brazed joints shall be evaluated in accordance with AWS C3.2M/C3.2.

b. Torch brazing shall be in accordance with AWS C3.4M/C3.4.

c. Induction brazing shall be in accordance with AWS C3.5M/C3.5.

d. Furnace brazing shall be in accordance with AWS C3.6M/C3.6. Furnace brazing of complex configurations such as heat exchangers and cold plates shall be subject to preproduction destructive testing to verify that the braze layer that extends beyond the fillet area is continuous and forms a uniform phase.

e. Brazing of critical and semi-critical aluminum alloy parts shall meet the requirements of AWS C3.7M/C3.7.

f. Fusion-welding or other high temperature processes that could affect the properties of brazed joints is restricted unless procedures, processes or methods can be demonstrated to prevent heating of the brazed joint to temperatures that can degrade the brazed joint.

5.3.8 Cold spray metallization. Cold spray metalized coatings may be used for non-structural dimensional restoration or galvanic protection and shall be in accordance with MIL-STD-3021. Surfaces shall be cleaned in accordance with MIL-STD-3021 and damage requiring dimensional restoration shall be prepared by blending to a repair-specific depth ratio as determined by the CEA. The surface preparation, powder handling, and cold spray process variables shall be set for each application or family of applications to ensure that cold spray operations are repeatable, consistent, and documented in accordance with MIL-STD-3021. The use of cold spray metallization on structural applications or any safety-of-flight critical or semi-critical applications is prohibited unless specifically approved by the CEA. The powdered raw materials used by cold spray (whether metallic, polymer, ceramic, or intermetallic) are a critical process variable, and shall be monitored or controlled in accordance with the standards approved by the CEA.

5.3.9 Soldering. Soldering shall not be used for structural applications.

5.3.10 Adhesive bonding. The SAE CMH-17 handbooks, MIL-HDBK-83377, or a CEA approved alternative shall be used for guidance on selection and use of adhesive bonding materials and processes. SAE AIR4938 gives training requirements for personnel performing bonding operations and SAE AIR6825 for composite/bonding structure inspectors.

5.3.11 Non-destructive inspection (NDI). NDI requirements shall be established and adhered to during system development, test, production, operation, and sustainment of the

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weapons systems. SAE AS6870 shall be used to identify and develop the key elements of an NDI program for the system being developed using industry standard NDI methodologies and techniques. When industry standard NDI methodologies and techniques cannot be utilized, alternate NDI methods may be utilized with authorization from the CEA. Any alternate NDI method shall be fully validated during the development phase before implementation in production. When developing inspections for critical structure, the structure material(s), design parameters (e.g., design strength, applied stresses, etc.), finishing systems and in-service NDI capabilities shall be considered. Unless otherwise authorized by the CEA, inspections shall be capable of detecting critical flaws and anomalies to a 90 percent probability of detection (POD) and a 95 percent confidence limit using production inspection hardware and personnel. Refer to MIL-HDBK-1823 guidance on executing probability of detection studies. NDI personnel performing these inspections shall meet the requirements of NAS 410 or an industry equivalent requirement which has been approved by the CEA.

5.4 Additive manufacturing. Additive manufacturing (AM) parts and materials have different material properties from equivalent wrought, sheet, plate, or cast parts and materials. To use AM parts and materials, a standard approach shall be taken to establish material allowables and to substantiate the qualification and certification of these AM parts. The AM process and material properties shall be in accordance with an Additive Manufacturing Procedure Specification (AMPS)/Procedure Qualification Record (PQR) as specified in AWS D20.1/D20.1M, or a Material Property Suite (MPS) as specified in NASA-STD-6030, or equivalent as approved by the CEA. The process and materials used in the systems shall be approved by the CEA and standardized to ensure repeatable quality. For example, the standardization of powders in a powder bed system is critical to repeatability. Powder characteristics, such as, powder size distribution, powder shape, manufacturing route (gas atomization versus plasma rotating electrode process), and powder composition, will have an effect on the resulting material properties of AM components. Post treatment of the AM parts, such as heat treatment, hot isostatic pressing (HIP), and surface finishing shall be defined for the specific application and material system to enhance or tailor the material properties. For reference, table V contains various industry standards for AM processes, feedstocks, definitions, policy, and products.

5.4.1 Mechanical property development and verification. In most cases, mechanical property allowables do not currently exist in published databases such as MMPDS or CMH-17 and therefore shall be developed and verified accordingly. Mechanical property design allowables shall include the effects of all sources of variation within the qualified process. This includes variation of qualification variables within specifications for feedstock, the build process, and post processing; see also the PQR in accordance with AWS D20.1/D20.1M or influence factors in accordance with NASA-STD-6030. The design allowable substantiation plan for AM products shall include assessment of the following:

a. Effects of feedstock variation within specification, such as chemical composition, particle size distribution, and powder reuse practices. Property development shall include feedstock variation to demonstrate adequate allowable stability for the production period.

b. Build process variability including build to build on one machine over time (through maintenance periods) and between machines (on a serial number and location basis) for products to be produced on multiple machines.

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c. Effects of surface finish, including the as deposited or modified surface roughness as applicable, for overhang angle, support structure remnants, etc.

d. Response to post-processing thermal treatments.

e. Witness sample equivalency for relevance to process control, including property sensitivity, e.g. whether room temperature tensile properties are sensitive to high temperature fatigue properties.

f. Correlation of specimen property distributions and component local property distributions in the final processed condition. Mechanical design properties shall be verified as representative of the part. Samples removed from parts shall include relevant features such as intersections and substrate heat affected zones in the case of integral substrates.

5.4.2 In-process correction. In-process corrections as defined in 3.11 shall not impair the ability of a part to perform its intended functions. The acceptability of any in-process corrections and the necessary validation testing and additional post-correction inspection to make such a determination shall be specified by the CEA. Such testing and inspection shall be in accordance with original requirements covered in 5.4.1 and 5.4.3. Material for these activities shall be produced using the same AMPS/PQR (AWS D20.1/D20.1M) or QMP (NASA-STD-6030) as required for the original part as described in 5.4. All in-process corrections shall be documented by the AM vendor in accordance with the CEA specified process. Pertinent details include but are not limited to part identifier, build layer, location in item, nature of initial nonconformance, nature of correction, results of additional validation testing, and results of post-correction inspection. Changes to previously qualified/approved AM process parameters shall not be classified as an “In-Process Correction.”

5.4.3 Quality assurance. Detailed quality assurance criteria for metallic and polymeric AM parts are contained within the metallic and polymeric AM sections below.

5.4.3.1 Process controls. Process controls shall be implemented to ensure that manufacturing processes are repeatable and yield parts free of defects detrimental to the design or intended service life and usage.

5.4.3.1.1 Material properties. The AM vendor shall develop a material mechanical properties database and associated process control reference distributions (as defined in 3.18) to enable the effective design of AM parts and development of robust manufacturing control limits. See also NASA-STD-6030.

5.4.3.1.2 Flaw size distributions. Unless otherwise approved by the CEA, the AM vendor shall develop a flaw size and type distribution database to enable the ability to conduct cost trade studies associated with various design approaches (see also AWS D20.1/D20.1M).

5.4.3.2 Witness coupons. Witness coupons, if required by the CEA, shall be manufactured in parallel and on the same build plate as the production AM parts. Witness coupon requirements, characterization procedures and acceptance criteria sufficient to ensure component integrity shall be established by the supplier and approved by the CEA.

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5.4.4 Metal additive manufacturing processes. Metallic additive manufacturing (M-AM), no matter the modality, shall be considered a process intensive manufacturing method. Because of this, qualification of M-AM processes as well as AM parts shall be managed similarly to other process intensive manufacturing methods like welding, casting, and composite layup. The qualification of M-AM processes and parts shall require the following documentation to validate consistent and repeatable process controls, material properties, dimensional accuracy, and part quality.

a. Feedstock specification. A feedstock specification to which modality-specific powders, wires, or other feedstock are qualified to ensure a controlled chemistry, manufacturing method, powder size distribution, wire diameter consistency, and other critical parameters. This document shall include feedstock handling and safety information, as well as guidance about feedstock management, conditioning, sampling, and blending/mixing, in the case of powders.

b. Final material specification. A specification to control the final chemistry and material properties of the as-built or post-processed M-AM material.

c. Machine qualification plan (MQP). Document containing all relevant equipment calibration records, dimensional accuracy test results, maintenance/requalification records to certify the equipment being used for manufacture of end items will be functionally equivalent to the equipment used to generate the material verification plan (MVP) data and associated allowables and operator training requirements.

d. Process control document (PCD). Document containing all relevant equipment settings, including control limits, for the specific modality and equipment being used to produce end items. This shall include, but is not limited to, build sequencing, build environmental conditions, local process adjustments (such as feed-forward control systems), distortion compensation, residual stress mitigation, and build interruption contingencies. This is required for both process qualification and part qualification.

e. Material verification plan (MVP). Document containing all relevant material test data, calculated material allowables, and post-processing information needed to validate the use of the selected M-AM modality and process. This is part specific.

f. Quality assurance plan (QAP). Document containing all relevant first article test (FAT) requirements, non-destructive inspection (NDI) requirements, lot acceptance requirements, defect size/location limits, repair/rework procedures, and dimensional accuracy validation to ensure the end item meets the form, fit, and function of the original design. This is part specific.

5.4.4.1 M-AM material allowables. Due to M-AM modalities being process intensive, material static allowables may be generated via three different methods depending on the needs of the end item and the material test data available. The method used for each end item shall be approved by the CEA.

a. MMPDS Volume II. Follow MMPDS guidance for M-AM statistical methods of C-Basis and D-Basis allowable calculation. This method is based on industry-wide values across multiple pieces of AM hardware, feedstock heat and lots and process

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lots. If the additive manufacturing supplier used by the CEA did not supply data for the generation of C- and D-Basis allowables, they shall provide data from their process that substantiates their inclusion.

b. Controlled local process. Follow MMPDS statistical methods for S/C/D-Basis allowable calculation but generate design values using only data generated from the exact hardware/feedstock/process used to produce the end item. Any minor variances between the test and production processes shall be accounted for and investigated for their impact on final material properties.

c. Validated process transfer. Follow MMPDS statistical methods for S/C/D-Basis allowable calculation, but the data used to generate this design value may be from different machines than the one producing the end items. In these cases, a validated transfer learning methodology approved by the CEA shall be used to adapt the M-AM process from one machine to another while maintaining equivalent statistical performance. If this approach is used, details of the validated transfer learning methodology shall be included in the MVP along with sufficient data to show the production configuration is statistically equivalent to the test configuration.

5.4.4.2 Quality assurance for metallic AM parts. Inspections of M-AM parts shall be implemented in accordance with AWS D20.1/D20.1M, or similar requirements to reliably detect defects. Inspection requirements shall be established based on characterization and documentation of the flaw types and orientations produced by the M-AM process and the demonstrated capability to detect these flaws throughout the part volume. Minimum inspection requirements are defined below based on criticality class (see 3.6). The inspection requirements will be included in the quality assurance plan (QAP).

5.4.4.2.1 Critical and semi-critical parts. Critical and semi-critical parts shall require 100 percent volumetric and surface inspection. Parts may be zoned for inspection based on durability and damage tolerance criteria where applicable.

a. Volumetric inspection. Volumetric inspections shall be performed by film or digital radiography in accordance with ASTM E1742/E1742M or computed tomography in accordance with ASTM E3375 and ASTM E1570 where appropriate. Inspection coverage and detection capability shall be qualified through demonstrations using representative quality indicators (RQIs) in accordance with ASTM E1817 or other appropriate standards as approved by the CEA. RQIs for CT process qualification shall represent the part material, heat treat condition, AM process and geometry. RQI flaw sizes and shapes shall be selected to quantify the smallest interpretable flaw size and include flaws positioned at locations representing regions of highest stress and regions of poorest image quality (i.e. region exhibiting lowest dose and image artifacts formed by geometry). The method for RQI design and measurement of CT performance metrics (including minimum voxel resolution, minimum contrast-to-noise values, and flaw sizing accuracy) shall be established and approved by the CEA. Ultrasonic inspection of wire feed direct energy deposition (DED) parts may be performed as an alternative to or in addition to radiography or computed tomography inspection, when approved by the CEA. Reference standard designs and use shall consider attenuation variability typical of DED builds. Ultrasonic inspection shall be performed in accordance with SAE AMS-STD-2154, ASTM E2375, SAE AMS2632,

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SAE AMS2631, or similar specification as approved by the CEA. Inspection processes, reference standards, surface preparation requirements shall be established, demonstrated, and approved by the CEA. Acceptance criteria shall be in accordance with 5.4.4.2.3 unless otherwise approved by the CEA.

b. Surface inspection. Surface finishing of all critical details shall be performed before surface inspection. Non-ferromagnetic and corrosion resistant stainless-steel alloy parts shall require penetrant testing penetrant testing (PT) of all critical surfaces in accordance with ASTM E1417/E1417M using a type 1 (fluorescent), sensitivity level 3 or better, a Method A (water washable) or D (hydrophilic emulsified) process to a procedure approved by the CEA. Ferromagnetic, non-corrosion resistant parts shall require magnetic particle testing (MPT) performed in accordance with ASTM E1444 using a process and procedure approved by the CEA.

5.4.4.2.2 Non-critical parts. The CEA shall define inspection and process control requirements for non-critical AM parts including close visual inspection and PT or MPT of all surfaces as a minimum, if appropriate. It should be noted that PT and MPT capability will be negatively impacted by the as-deposited surface condition. PT on surfaces in other than the fully machined condition shall be performed when required by the CEA to identify gross defects (e.g. base plate separation, heat treat cracking, etc.). PT on as-deposited surfaces of non-ferromagnetic and corrosion-resistant steel alloys, shall be performed in accordance with ASTM E1417/E1417M using a type 1 (fluorescent), sensitivity Level 2 or better, a Method A (water washable) process to a procedure approved by the CEA. MPT on as-deposited surfaces of non-corrosion resistant ferromagnetic steel alloys shall be performed in accordance with ASTM E1444 using a process and procedure approved by the CEA.

5.4.4.2.3 Acceptance criteria.

a. Penetrant, radiography, and computed tomography acceptance criteria. The acceptance criteria of AWS D20.1/D20.1M shall apply to radiography and penetrant inspection of metallic AM parts. Class A requirements shall apply to Critical AM parts, Class B requirements shall apply to Semi-Critical AM parts and Class C requirements shall apply to non-critical parts. Alternate acceptance criteria may be established as approved by the CEA.

b. Ultrasonic inspection acceptance criteria. The acceptance criteria of SAE AMS 2154 shall apply to ultrasonic inspection of metallic AM parts. Class AAA shall apply to Critical Parts; Class AA shall apply to Semi-Critical Parts and Class A shall apply to Non-Critical Parts. Alternate acceptance criteria may be established as approved by the CEA.

5.4.4.2.4 Inspection capability demonstration. Inspections shall provide the capability to reliably detect flaws of the type and size defined by the acceptance criteria in 5.4.4.2.3. Detection capability demonstrations shall be performed for inspection processes where quantitative demonstration data or where method specific performance demonstration requirements do not exist. Capability demonstrations shall be established and approved by the CEA. Inspection capability for AM parts shall be demonstrated using parts representing the AM process, part geometry, surface condition, material and heat treat condition. When a probability of detection (POD) (see 5.3.11) demonstration is required, representative defects shall be embedded to

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statistically quantify the inspection capability for the flaw types and locations of concern. MIL-HDBK-1823 should be used to guide the experimental design and statistical analysis.

5.4.5 Polymeric additive manufacturing processes. Polymeric additive manufacturing (P-AM) is a process intensive manufacturing method. Qualification of P-AM components shall be managed similarly to other process intensive manufacturing methods where material-process-property relationships shall be taken in to account. Government and industry guidance provide best practices for qualification, design, and quality assurance of builds in ISO/ASTM 52901, ASTM F3091, ISO/ASTM 52936-1 and NASA STD-6030. The qualification of P-AM parts/processes shall require the following documentation to validate consistent and repeatable material properties, dimensional accuracy, and part quality. Government and industry guidance provide examples and best practices and guidance for material (SAE AMS7101, ISO/ASTM 52903-1, ISO/ASTM 52903-2, ASTM F3606) and process (AF TO 34A-3-1, NAWCADLKE-BL20000-DDR-001, SAE AMS7100/1, ASTM F3529, NASA STD-6030) specifications for some additive processes.

a. Feedstock specification. Specification to which modality-specific powders, filaments, or other feedstock are qualified to ensure a controlled chemistry, manufacturing method, powder size distribution, filament diameter consistency, moisture content, and other critical parameters. This document shall include feedstock handling and safety information, as well as guidance about feedstock management, conditioning, sampling, and blending/mixing.

b. Final material specification. Specification to control the final chemistry and material properties of the as built or post-processed AM material.

c. Material verification plan (MVP). Document containing all relevant material test data, calculated material allowables, and post-processing information needed to validate the use of the selected P-AM modality and process.

d. Machine qualification plan (MQP). Document containing all relevant equipment calibration records, dimensional accuracy test results, maintenance/requalification records to certify the equipment being used for manufacture of end items will be functionally equivalent to the equipment used to generate the MVP data and associated allowables and operator training requirements.

e. Process control document (PCD). Document that defines and controls the process settings to ensure consistent production. This includes equipment settings, including control limits, for the specific modality and equipment being used to produce end items. This shall include, but is not limited to, build sequencing, build environmental conditions, local process adjustments (such as feed-forward control systems), distortion compensation, residual stress mitigation, and build interruption contingencies.

f. Quality assurance plan (QAP). Document containing all relevant first article test (FAT) requirements, non-destructive inspection (NDI) requirements, defect size/location limits, repair/rework procedures, and dimensional accuracy validation to ensure the end item meets the form, fit, and function of the original design.

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5.4.5.1 P-AM material allowables. In addition to these requirements, the anisotropy inherent to additive manufacturing processes for non-metallic P-AM parts shall be evaluated and data developed along different build axes as appropriate. Specific print parameters that will impact material properties such as print bed temperature, print speed, laser power, extrusion temperature, etc. shall also be evaluated and controlled. Print design settings that will impact material properties such as layer height, number of contours, infill density and geometry, etc. shall be evaluated and controlled. Test methods for property generation shall be approved. Material allowables shall be generated from test data following guidelines for A-basis or B-basis allowables in accordance with the SAE CMH-17 (Composite Material Handbook, Volumes 1-6). B-basis values are generally accepted as the standard for polymer/composite allowables. Alternate statistical methods are acceptable for generating material data providing evidence of material property and process stability with engineering authority approval.

5.4.5.2 Quality assurance for polymeric AM part.

a. Critical components. Quality assurance requirements for polymeric AM critical parts shall be established and approved by the CEA and shall be sufficient to ensure the parts have no defects that adversely affect the performance of the part.

b. Non-critical components. The CEA shall define inspection and process control requirements for non-critical P-AM parts. All non-critical P-AM parts shall require close visual inspection for previously identified unacceptable defects and meet dimensional requirements. Additional inspection requirements shall be developed and agreed upon in accordance with 5.3.11.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This design standard provides design criteria that provide timely and comprehensive consideration during systems design of the limitation of materials and processes and of the lessons learned over the years from military-unique operational systems worldwide. The standard employs military-unique design criteria and considerations that exceed commercial design practices in order to meet the military rigors and environment that the operational systems encounter. The use of this document will result in more durable systems in operational service. It should be used in conjunction with MIL-STD-1568 in selection of materials and processes that will meet the requirements of the system being designed in accordance with MIL-STD-1530, MIL-STD-1798, and MIL-STD-3024 or equivalent integrity program documents.

6.2 Acquisition requirements. Acquisition documents should specify the title, number, revision letter and date of this standard.

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6.3 Associated Data Item Descriptions (DIDs). This standard has been assigned an Acquisition Management Systems Control number authorizing it as the source document for the following DIDs. When it is necessary to obtain the data, the applicable DID must be listed on the Contract Data Requirements List (DD Form 1423).

Paragraph #	DID Title	Applicable DID
4.4, 4.4.1, 4.4.2	Program-Unique Material or Process Specification	DI-MFFP-82119

The above DID was current as of the date of this standard. The ASSIST database should be researched at (http://quicksearch.dla.mil) to ensure that only current and approved DIDs are cited on the DD Form 1423.

6.4 Subject term (key word) listing.

Aircraft materials and processes
Composites
M&P lessons learned
Material properties
Metals
Non-metals
Additive manufacturing

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the vast extent of the changes.

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Table I. Restricted alloys, heat treatment conditions and product forms.

Aluminum Alloys 1/	Form/Heat Treat Condition	Reason	Exceptions Allowed 2/
2014	All	3/	Wheel Forgings per 1/
2020	All	3/	
2024	T6, T8	3/	T8 Rod, Bar, Plate or Sheet per 1/
2024	T3, T4	3/ 4/	
2XXX	T3, T4	3/	
7XXX	T6, T76	3/ 7/	USARMY & USAF Extrusions under 0.250" thick. USNAVY no exceptions for marine environments 4/
Steels	**Heat Treat Condition**	**Reason**	**Exceptions Allowed 2/**
4130, 4140, 4330V, 4340, 300M, etc.	Above 180 ksi UTS	3/ 5/ 6/ 7/	4340 & 300M USAF Landing Gear Components
H-11, D6AC	All	3/ 5/ 6/	
Ni Maraging Steel Alloys 250, 300, 350	All	3/ 5/	
HP9-4-30	All	3/	
AerMet 100	H875 & H900	3/	
Stainless Steels	**Heat Treat Condition**	**Reason**	**Exceptions Allowed 2/**
303, 303Se		5/	
400 Series Stainless Steels (except 431 & 440C)	Tempered between 700 ºF & 1100 ºF (150-180 ksi UTS)	3/ 5/	
431	All	5/	
440C	All	3/	
15-5PH	Aged below 1000 ºF (USAF & USARMY) Aged below 1025 ºF (USN/USMC)	3/	
17-4PH	Aged below 1000 ºF (USAF & USARMY) Aged below 1025 ºF (USN/USMC)	3/	Castings aged at 935 ºF, Fasteners in the H950 condition and Springs in the CH900 Condition
PH13-8Mo	Aged below 1000 ºF	3/	
Custom 450, 455	Aged below 1000 ºF	3/	
Custom 465	Aged below 1000 ºF	3/ 5/	
17-7PH	Above 200 ksi UTS	3/ 5/	Springs in the CH900.
PH15-7Mo	All except CH900	3/	CH900

1/ In all cases, the minimum SCC resistance of aluminum alloys in any heat treatment condition, exposed to SCC environments shall be 25 ksi or higher as determined in accordance with MIL-STD-1568.

2/ See section 4.2 for acceptable use of in-service alloys and heat treatment conditions for parts, spares, etc.

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3/ Low resistance to stress corrosion cracking.
4/ Poor exfoliation corrosion resistance.
5/ Low fracture toughness.
6/ Extreme sensitivity to hydrogen embrittlement during manufacture and rework.
7/ Newer alloys available with much improved properties, less in-service issues, and overall lower lifetime costs.

TABLE II. O-rings.

System	Materials Specification	Applicable Drawing
Fuel	SAE AMS-P-5315 (NBR)	SAE AS29512, SAE AS29513
	MIL-DTL-25988 (Fluorosilicone)	MIL-R-25988/1, /2, /3, & /4
	SAE AMS7260 (NBR)	
	SAE AMS7271 (NBR)	
Lubrication	MIL-DTL-25988 (Fluorosilicone)	MIL-R-25988/1, /2, /3, & /4
	SAE AMS7273 (Fluorosilicone) SAE AMS7276 (Fluorocarbon) SAE AMS7259 (Fluorocarbon) SAE AMS3325 (Fluorosilicone) SAE AMS3327 (Fluorosilicone) SAE AMS7287 (Fluorocarbon)	SAE AS3208, SAE AS3209, SAE AS3581
	SAE AMS-R-7362 (NBR)	SAE AS29561
Hydraulic	SAE AMS-P-83461 (NBR)	SAE AS83461/1, SAE AS83461/2
	MIL-PRF-25732 (NBR)	SAE AS28775
	SAE AMS-P-5510 (NBR)	SAE AS28778

TABLE III. Other molded parts, sheets, strips, and extruded shapes.

Specification	Material
MIL-PRF-6855	NBR, CR and SBR 1/
SAE AMS-R-7362	NBR 1/
MIL-DTL-25988	Fluorosilicone
SAE AMS7276, SAE AMS7259, SAE AMS3216 & SAE AMS3218	Fluorocarbon elastomer
SAE AMS-R-83285	Ethylene-propylene
MIL-DTL-83397	Polyurethane
SAE AMS3384 SAE AMS7287	Low Temp Fluorocarbon elastomer High Temp Fluorocarbon elastomer
A-A-59588, SAE AMS7267, SAE AMS3304, SAE AMS3305 & SAE AS9386	Silicone

1/ NBR = Acrylonitrile Butadiene Rubber, CR = Chloroprene Rubber, SBR = Styrene – Butadiene Rubber.

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TABLE IV. Potting compounds. 1/

Specification	Material
MIL-PRF-8516	Polysulfide
A-A-59877	Epoxy
MIL-PRF-23586	Silicone
MIL-M-24041	Polyurethane
SAE AS81550	Silicone

1/ Materials other than those meeting the above specifications shall only be used if approved by the CEA.

TABLE V. Additive manufacturing specifications.

Issuing Org.	Spec/STD #	Title	Type
ISO/ASTM	52900:2021	Additive Manufacturing - General Principles - Fundamentals and Vocabulary	Definitions
ISO/ASTM	52903-1-20	AM Material Extrusion Plastic Feedstock	Feedstock
SAE	AMS7002	Nickel Alloy, Corrosion and Heat-Resistant, Powder for Additive Manufacturing, 62Ni - 21.5Cr - 9.0Mo - 3.65Nb	Feedstock
SAE	AMS7006	Nickel Alloy, Corrosion- and Heat-Resistant, Powder for Additive Manufacturing 52.5Ni - 19Cr - 3.0Mo - 5.1Cb (Nb) - 0.90Ti - 0.50Al - 18Fe	Feedstock
SAE	AMS7008	Nickel Alloy, Corrosion and Heat-Resistant, Powder for Additive Manufacturing, 47.5Ni - 22Cr - 1.5Co - 9.0Mo - 0.60W - 18.5Fe	Feedstock
SAE	AMS7012	Precipitation Hardenable Steel Alloy, Corrosion and Heat-Resistant Powder for Additive Manufacturing 16.0Cr - 4.0Ni - 4.0Cu - 0.30Nb	Feedstock
SAE	AMS7013	Nickel Alloy, Corrosion and Heat-Resistant, Powder for Additive Manufacturing, 60Ni - 22Cr - 2.0Mo - 14W - 0.35Al - 0.03La	Feedstock
SAE	AMS7014	Titanium Alloy, High Temperature Applications, Powder for Additive Manufacturing, Ti - 6.0Al - 2.0Sn - 4.0Zr - 2.0Mo	Feedstock
SAE	AMS7015	Titanium 6-Aluminum 4-Vanadium Powder for Additive Manufacturing	Feedstock
SAE	AMS7017	Titanium 6 - Aluminum 4 - Vanadium Powder for Additive Manufacturing, Extra Low Interstitial (ELI)	Feedstock
SAE	AMS7018	Aluminum Alloy Powder 10.0Si - 0.35Mg	Feedstock
SAE	AMS7020	Aluminum Alloy Powder 7.0Si - 0.55Mg - 0.12Ti	Feedstock
SAE	AMS7021	Additive Manufacturing, 15.0Cr - 4.5Ni - 3.5Cu - 0.30Nb	Feedstock
SAE	AMS7023	Gamma Titanium-Aluminide Powder for Additive Manufacturing Ti -48Al - 2Nb - 2Cr	Feedstock
SAE	AMS7025	Metal Powder Feedstock Size Classifications	Feedstock
SAE	AMS7026	Titanium Ti-5553 (Ti - 5Al - 5Mo - 5V - 3Cr) Powder for Additive Manufacturing	Feedstock
SAE	AMS7033	Aluminum Alloy Powder 4.6Cu - 3.4Ti - 1.4B - 0.75Ag - 0.27Mg	Feedstock
SAE	AMS7035	Precipitation Hardenable Steel Alloy, Corrosion and Heat-Resistant, Powder for Binder Jet Additive Manufacturing, 16.0Cr - 4.0Ni - 4.0Cu - 0.30Nb	Feedstock
SAE	AMS7037	Steel, Corrosion and Heat-Resistant, Powder for Additive Manufacturing, 17Cr - 13Ni - 2.5Mo (316L)	Feedstock
SAE	AMS7101/2	Fused Filament Fabrication, Type 1, Class 1, Form 1, Grade 0, Natural Color Material for	Feedstock
SAE	AMS7101A	Fused Filament Fabrication, Material for	Feedstock

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TABLE V. Additive manufacturing specifications – Continued.

Issuing Org.	Spec/STD #	Title	Type
SAE	AMS7002A	Process Requirements for Production of Metal Powder Feedstock for Use in Additive Manufacturing of Aerospace Parts	Feedstock Process
SAE	AMS7031	Batch Processing Requirements for the Reuse of Used Powder in Additive Manufacturing of Aerospace Parts	Feedstock Process
ASTM	F3490	Standard Practice for Additive Manufacturing - General Principles - Overview of Data Pedigree	Policy
ASTM	F3529	Guide for Additive Manufacturing – Design - Material Extrusion of Polymers	Policy
ASTM	F3572	Standard Practice for Additive Manufacturing - General Principles - Part Classifications for Additive Manufactured Parts Used in Aviation	Policy
ISO/ASTM	52910:2021	Additive manufacturing - Design - Requirements, guidelines and recommendations	Policy
ISO/ASTM	52920	Additive manufacturing - Qualification principles - Requirements for industrial additive manufacturing processes and production sites	Policy
ISO/ASTM	52930:2021	Additive manufacturing - Qualification principles - Installation, operation and performance (IQ/OQ/PQ) of PBF-LB equipment	Policy
ISO/ASTM	52942	Additive manufacturing - Qualification principles - Qualifying machine operators of laser metal powder bed fusion machines and equipment used in aerospace applications	Policy
NASA	6030	Additive Manufacturing Requirements for Spaceflight Systems	Policy
ASTM	F3187	Standard Guide for Directed Energy Deposition of Metals	Process
ASTM	E3166	Standard Guide for Nondestructive Examination of Metal Additively Manufactured Aerospace Parts After Build	Process
ASTM	E3353	Standard Guide for In-Process Monitoring Using Optical and Thermal Methods for Laser Powder Bed Fusion	Process
ASTM	F2971	Standard Practice for Reporting Data for Test Specimens Prepared by Additive Manufacturing	Process
ASTM	E3637	Standard Guide for Additive Manufacturing of Metal - Finished Part Properties - Methods for Relative Density Measurement	Process
ASTM	F3301	Standard for Additive Manufacturing - Post Processing Methods - Standard Specification for Thermal Post-Processing Metal Parts Made Via Powder Bed Fusion	Process
ASTM	F3413	Guide for Additive Manufacturing – Design - Directed Energy Deposition	Process
AWS	D20.1/D20.1M	Specification for Fabrication of Metal Components using Additive Manufacturing	Process
ISO/ASTM	52903-2	AM Material Extrusion Process Equipment	Process
NASA	6033	Additive Manufacturing Requirements for Equipment and Facility Control	Process
SAE	AMS7003A	Laser Powder Bed Fusion Process	Process
SAE	AMS7005	Wire Fed Plasma Arc Directed Energy Deposition Additive Manufacturing Process	Process
SAE	AMS7007	Electron Beam Powder Bed Fusion Process	Process
SAE	AMS7010A	Laser Directed Energy Deposition Additive Manufacturing Process (L-DED)	Process
SAE	AMS7022	Binder Jet Additive Manufacturing (BJAM) Process	Process
SAE	AMS7027	Electron Beam Directed Energy Deposition-Wire Additive Manufacturing Process (EB-DED-Wire)	Process
SAE	AMS7032	Machine Qualification for Fusion-Based Metal Additive Manufacturing	Process
SAE	AMS7100	Fused Filament Fabrication, Process Specification for	Process

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TABLE V. Additive manufacturing specifications – Continued.

Issuing Org.	Spec/STD #	Title	Type
SAE	AMS7100/1	Fused Filament Fabrication, Stratasys 900mc plus with Type 1, Class 1, Form 1, Grade 0 Natural Color Material for AMS7100/1	Process
SAE	AMS7100/2	Fused Filament Fabrication Process - Stratasys Fortus 900mc Plus with Type 1, Class 1, Form 1, Grade 0 Natural Color Material for AMS7100/2	Process
SAE	AMS7101/1	Fused Filament Fabrication, Type 1, Class 1, Form 1, Grade 0, Natural Color Material for AMS7101/1	Process
ISO/ASTM	52904	Additive Manufacturing - Process Characteristics and Performance: Practice for Metal Powder Bed Fusion Process to Meet Critical Applications	Process
ISO/ASTM	52941	Additive manufacturing - System performance and reliability - Acceptance tests for laser metal powder-bed fusion machines for metallic materials for aerospace application	Process
ASTM	F3055	Standard Specification for Additive Manufacturing Nickel Alloy (UNS N07718) with Powder Bed Fusion	Product
ASTM	F3056	Standard Specification for Additive Manufacturing Nickel Alloy (UNS N06625) with Powder Bed Fusion	Product
ASTM	F3091	Standard Specification for Powder Bed Fusion of Plastic Materials	Product
ASTM	F3184	Standard Specification for Additive Manufacturing Stainless Steel Alloy (UNS S31603) with Powder Bed Fusion	Product
ASTM	F3213	Standard for Additive Manufacturing - Finished Part Properties - Standard Specification for Cobalt-28 Chromium-6 Molybdenum via Powder Bed Fusion	Product
ASTM	F3302	Standard for Additive Manufacturing - Finished Part Properties - Standard Specification for Titanium Alloys via Powder Bed Fusion	Product
ASTM	F3318	Standard for Additive Manufacturing - Finished Part Properties - Specification for AlSi10Mg with Powder Bed Fusion - Laser Beam	Product
SAE	AMS4999A	Titanium Alloy Direct Deposited Products 6Al-4V Annealed	Product
SAE	AMS7000A	Laser-Powder Bed Fusion (L-PBF) Produced Parts, Nickel Alloy, Corrosion- and Heat-Resistant, 62Ni - 21.5Cr - 9.0Mo - 3.65Nb Stress Relieved, Hot Isostatic Pressed and Solution Annealed	Product
SAE	AMS7004	Titanium Alloy Preforms from Plasma Arc Directed Energy Deposition Additive Manufacturing on Substrate Ti-6Al-4V Stress Relieved	Product
SAE	AMS7011	Electron Beam-Powder Bed Fusion (EB-PBF) Produced Preforms and Parts Titanium Alloy, 6Al - 4V Hot Isostatically Pressed	Product

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CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - AS
Air Force - 20

Review activities:
Army - AR, AV, MR
Navy - CH, EC, MC, SA
Air Force - 11

Preparing activity:
Navy-AS
(Project MFFP-2023-009)

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